Hydropower Collegiate Competition (HCC) 2027

Initial Design Report

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DISCLAIMER

This report was prepared by students as part of a university course requirement. While considerable effort has been put into the project, it is not the work of licensed engineers and has not undergone the extensive verification that is common in the profession. The information, data, conclusions, and content of this report should not be relied on or utilized without thorough, independent testing and verification. University faculty members may have been associated with this project as advisors, sponsors, or course instructors, but as such they are not responsible for the accuracy of results or conclusions.

EXECUTIVE SUMMARY

Team HCC 2027 represents Northern Arizona University in the Hydropower Collegiate Competition sponsored by the U.S. Department of Energy. The project focuses on developing hydropower design that meets the competition's technical requirements while considering energy generation, cost, environmental impact, and implementation. The competition requires the team to evaluate the feasibility of its design and communicate its engineering decisions through technical reports, presentations, and other required deliverables.

The team is evaluating potential non-powered dam sites for hydropower development using MATLAB to compare site characteristics and identify promising locations. The evaluation considers power capacity, annual energy output, variations in flow and head, proximity to electrical infrastructure, and local energy costs. These factors provide a basis for comparing sites and determining which locations offer the best potential for a hydropower installation. The initial screening process evaluated approximately 2600 candidate dams and narrowed the selection to leading sites for further analysis. New Croton Reservoir Dam, Little Falls Dam, and Birch Hill Dam were identified as the three leading candidates.

The technical design process includes researching existing hydropower systems, developing mathematical models, and evaluating turbine design choices based on site specific conditions. The relationship between flow rate, hydraulic head, and turbine efficiency is used to estimate available power and guide preliminary turbine selection. Additional analysis considers environmental flow requirements, electrical infrastructure, and other constraints that may affect the feasibility of the proposed system.

The initial design report establishes the project's engineering requirements and documents the research, calculations, and concept selection methods that will guide further development. Project success will be evaluated through the ability to meet the established requirements, support design decisions with engineering calculations, and complete the required competition deliverables. Further analysis and design refinement will be necessary to verify the feasibility and expected performance of the selected concepts. The project also considers installation costs, electrical grid connection, environmental requirements and the need to maintain existing dam operations. Further analysis and design refinement will be necessary to verify the feasibility of the selected concepts.

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1 BACKGROUND

Northern Arizona University's Mechanical Engineering Capstone team is participating in the 2027 Hydropower Collegiate Competition (HCC). The project focuses on evaluating existing non-powered dams (NPDs) for potential conversion into hydroelectric facilities. The team will consider engineering feasibility, economic costs, environmental impacts, and community engagement throughout the project. This section introduces the project's objectives, outlines the course and competition deliverables, and establishes the success metrics used to evaluate the team's progress.

1.1 Project Description

Non-powered dams (NPDs) throughout the United States represent untapped potential for renewable energy production. Converting these dams into hydroelectric facilities allows electricity to be generated using existing infrastructure, reducing the need for new construction. The 2027 Hydropower Collegiate Competition (HCC), sponsored by the U.S. Department of Energy (DOE), challenges university teams to evaluate NPDs capable of producing between 1 and 10 megawatts (MW) of electricity[1]. Site selection must consider hydraulic head, flow rate, economic feasibility, environmental impacts, and access to existing infrastructure. The competition also emphasizes community outreach to educate the public and connect students with professionals in the hydropower industry.

Northern Arizona University's Mechanical Engineering Capstone team will evaluate potential dam sites using mathematical modeling, engineering calculations, and decision matrices. The team will use MATLAB and Excel to estimate power generation, analyze hydraulic conditions, and compare potential locations. Once a site is selected, the team will develop preliminary designs using computer-aided design (CAD) software and fluid simulations to evaluate the proposed hydropower system. In addition to engineering analysis, the team plans to interview industry professionals and water management personnel to gain insight into hydropower development. Educational outreach events at local middle and high schools will also be conducted, using pre- and post-event surveys to measure student understanding of hydropower.

The team must also manage project expenses while meeting academic and competition requirements. The preliminary team budget is $15,000, which will support project materials, prototype development, travel, and outreach activities. The team will evaluate its expected expenses against available funding to determine whether additional financial support is needed. If necessary, additional funding will be pursued through university resources, industry sponsorships, and fundraising opportunities. Maintaining a budget will help the team allocate resources effectively and complete the project within its financial constraints.

1.2 Deliverables

The project deliverables are divided into two categories: course deliverables and competition deliverables. Course deliverables focus on documenting the engineering design process and demonstrating the team's progress throughout the academic year. Competition deliverables focus on site selection, hydropower system design, and community outreach. Together, these deliverables establish the major milestones required to complete the project.

1.2.1 Course Deliverables

The Mechanical Engineering Capstone course requires the team to demonstrate its progress through presentations, technical reports, engineering designs, and project documentation. The next major deliverable is Presentation 3, which will cover the team's research, engineering requirements, mathematical modeling, site selection, and preliminary design concepts. The presentation will allow the team to receive feedback and identify areas requiring further development.

As the project progresses, the team will continue developing mathematical models, CAD drawings, and engineering calculations to support the proposed hydropower system. Future deliverables will include updated technical reports, detailed designs, and prototype development and testing as required by the course. The project will conclude with final engineering documentation and a presentation at NAU's Engineering Symposium, where the team will present its design process, calculations, and results.

1.2.2 Competition Deliverables

The 2027 Hydropower Collegiate Competition includes three main challenges: Siting, Design, and Community Connections. Each challenge requires the team to evaluate different aspects of hydropower development and document its findings.

The Siting Challenge focuses on identifying and evaluating existing non-powered dams for potential hydropower conversion. The team will compare candidate sites based on hydraulic conditions, power generation potential, environmental impacts, and economic feasibility. The results will be supported by engineering calculations, site data, decision matrices, and a site selection report.

The Design Challenge focuses on developing a hydropower system for the selected dam. The team will evaluate turbine selection, penstock design, hydraulic losses, and overall system performance. CAD drawings, mathematical models, and fluid simulations will support the design and provide technical justification for the proposed system.

The Community Connections Challenge focuses on connecting with industry professionals and educating local communities about hydropower. The team plans to conduct professional interviews and educational outreach events at middle and high schools. Interviews, participation records, and pre- and post-event surveys will document the results of these activities.

The team will also prepare a technical poster and quick pitch to summarize its engineering work and outreach efforts. The optional Build and Test Challenge may provide an additional opportunity to demonstrate and evaluate a physical prototype. All competition deliverables will follow the official HCC 2027 requirements and submission schedule.

1.3 Success Metrics

The success of the HCC 2027 project will be determined by the team's ability to meet competition requirements, develop a feasible hydropower system, and complete the required deliverables. In addition to meeting these requirements, the team aims to improve system efficiency, reliability, and overall performance. The following metrics will be used to evaluate the project's success.[1]

Power Generation: Identify a non-powered dam capable of producing between 1 and 10 MW of electricity. Potential power generation will be calculated using available hydraulic head, flow rate, and system efficiency.

System Efficiency: Develop a hydropower design that minimizes energy losses and maximizes electrical output. Turbine efficiency, penstock head losses, and overall system performance will be evaluated to determine whether the design meets the team's engineering requirements.

Reliability and Safety: Design a system that can operate reliably under expected conditions while maintaining the dam's existing functions. Mechanical components will be evaluated using engineering calculations, appropriate factors of safety, and expected operating loads.

Economic Feasibility: Evaluate the costs associated with developing and maintaining the proposed hydropower facility. The team will consider equipment costs, available infrastructure, and expected power generation to determine the feasibility of the project.

Environmental Impact: Minimize the effects of hydropower conversion on surrounding ecosystems and existing water operations. The design will consider fish passage, sediment transport, and environmental regulations applicable to the selected site.

Engineering Design: Develop a preliminary system supported by mathematical models, CAD drawings, and fluid simulations. These methods will be used to evaluate hydraulic performance, mechanical requirements, and the feasibility of the proposed design.

Community Engagement: Conduct interviews with industry professionals and educational outreach events for middle and high school students. Success will be measured through participation records, professional feedback, and pre- and post-event surveys evaluating student understanding of hydropower.

Competition Performance: Complete all required competition deliverables and aim to place in at least one competition category. The team will track progress through project milestones, submission deadlines, and feedback from faculty advisors and competition judges.

These metrics will be verified through engineering calculations, CAD modeling, fluid simulations, and custom MATLAB programs. These methods will provide quantifiable results for power generation, hydraulic losses, efficiency, and mechanical performance. The team will compare these results against its engineering requirements to identify areas for improvement and evaluate the overall success of the project.

2 REQUIREMENTS

This section will cover the customer requirements listed for the siting challenge and go into what they mean in quantifiable terms. It will also be presented and talk about the HoQ that was developed for the siting challenge. While there are other deliverables for the project as a whole the team has no chosen which track to go down for those deliverables

2.1 Customer Requirements (CRs)

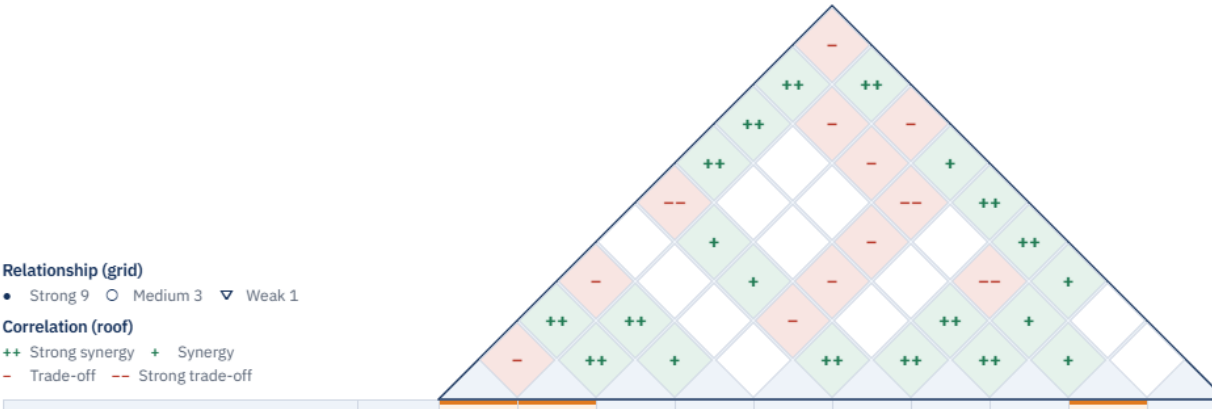
1. High-level costs: The costs of labor and installation as well as the cost of the main components such as a generator, turbine, shaft, penstock gates, modified penstock piping, grid hookup costs, etc
2. Resources and generation availability: The dam must have the resources to generate the required power and not always pull all the water from the river.
3. Dam safety and geotechnical data: The dam must be able to contain the extra stress of the electrical addiction as well as not pose an issue from earthquakes.
4. Access to transmission/grid integration: Ensure that it will be able to connect to a grid without adding significant cost and complexity
5. Transportation access: Make sure that laborers and maintenance workers can access the space to completer their work
6. Environmental factors e.g. fish passage, sensitive species: Protect the lands surrounding the dam and ensure that animals that depend on the river can pass through
7. Cultural factors e.g. historical landmarks: The cultures surrounding the river and the reservoir may hold it in high regard, ensure that it is not an issue to add anything to the dam
8. Social metrics: Neighboring towns or peoples may take issue with the alteration or the noise from its operation. Make sure the neighbors approve of the alteration.
9. Operation and maintenance requirements: The operation of the dam will require personal to run and maintain it safely.
10. Agricultural effects: Ensure that farms that rely on the reservoir will keep their water rights and access.
11. Triple bottom line assessment of options (economic environments and social outcomes): This is covered by earlier requirements, just in review.
12. Opportunities to reduce the cost of adding power to existing infrastructure: This is an iteration based requirement.

2.2 Engineering Requirements (ERs)

The engineering requirements are based off of the customer requirements but altered for our understanding and quantifiability.

1. Installation cost: Total cost of the installation of the dam should be between 8-10 million dollars.
2. Max head: For turbine operation, head must be between 2-400 m of head, which is the main driver of pressure in the system.
3. Required flow: 0.5-50 m^3/s of water through the turbine, with a smaller head there will be need of higher flow rate to achieve desired power
4. Reservoir capacity: How much water will be held behind the dam and therefore potential energy, 1-10 million m^3
5. Local electric prices: The cost of utilities native to the area have a large effect on productivity and need, >13 cents/kWh
6. Cost recoup time: With a limited lifespan for the components this needs to be low in order to be feasible, 1 year
7. Infrastructure distance: Large distance can quickly become the largest expenditure, 1 mile ideally to substation
8. Site accessibility: While not as expensive as electric workers need to be able to get there, <10 miles
9. Ecological impact: Try to disturb as little of the surroundings as possible, reservoir included, 15,000 acres
10. Energy production: Given by the department of energy on how much the dam needs to be able to produce 1-10MW

2.3 House of Quality (QFD)



Customer needs What customers want, in their own words		Importance (1 to 5)	instalation cost	max head	required flow	resevoir capacity	local electric prices	cost recup time	infrastructure distance	site accesibility	ecological impact	energy production
Direction of improvement			▼	▲	▼	▲	▲	▼	▼	▲	▼	◎
resource and generation ability	×	5	○	●	●	●	●	●	○	▽	●	●
enviromental factors	×	4	▽	○	○	●	●	▽	○	○	●	○
dam safety	×	3	○	●	○	●					○	
access to transmission and grid integ	×	5	●				○	●	●	○	▽	▽
transportation access	×	2	●					○	○	●	○	
cultural factors	×	1	▽	▽	●	○			▽	○	●	
operation and maintanance requirem	×	3	○	○	●	▽		●	▽	○		○
agricultural effects	×	3	▽	▽	○	○			▽	▽	●	▽
social metrics (will the people approv	×	2	○	○	○	▽	○			○	○	▽
high level costs	×	5	●	●	▽	▽	○	○	●	○	▽	●

Figure 1: House of Quality

The house of quality shown above shows the relationship between all of the differing engineering requirements as well as how those requirements corelate with the customer requirements.

Target value	8-10 million	50M	24-60 M^3/s	1-10 million m^3	>13 cents/kWh	1 year	1 mile	10 miles	<15000 acres disturbed	35,000 MWh/yr
Importance score	155	148	122	130	117	142	130	86	148	121
Relative weight	12%	11%	9%	10%	9%	11%	10%	7%	11%	9%
Priority rank	#1	#2	#7	#5	#9	#4	#5	#10	#2	#8

Figure 2: Bottom of House of Quality

Figure 2 shows us which engineering requirements should be considered first and which do not need much consideration. As this is a living document it can change if needed to alter some of the weights for the customer requirements or if other engineering requirements come into play that have yet to be listed.

3 Research Within Your Design Space

This section reviews the benchmarking, research, and mathematical models that our team completed as part of our initial design. It covers design flow vs cost estimations, system efficiency quantification, and advanced turbine selection metrics. These analyses give the foundation for concept selection and generation, along with a framework for future design development.

3.1 Benchmarking

3.1.1 Benchmarking Overview

This section presents three system-level benchmarks of non-powered dam (NPD) conversions in the 1–10 MW range. The team considers these projects state-of-the-art because they are recent, operating conversions that generate power from existing dams without changing the dam's original purposes. The team uses these projects for an overview of the conversion process: the design choices developers made to get the most value from an existing dam, and the challenges they faced along the way. The team also benchmarks three sub-systems (a turbine, a penstock and gate, and a fish passage) from other operating projects for a detailed look at proven components currently in use.

3.1.2 System-Level Benchmarking

3.1.2.1 Mahoning Creek Hydroelectric Project

The team benchmarked the Mahoning creek Hydroelectric Project in Armstrong County, Pennsylvania, as a complete conversion of a non-powered dam. Enduring Hydro converted the dam to a 6 mega-watt (MW) hydropower plant designed to produce more than 20,000 mega-watt-hour (MWh) per year. The team selected this project because its capacity falls within the competition's 1–10 MW range and its design applies directly to the team's own conversion. A 1,090-ft, 10-ft diameter buried steel penstock carries water from the dam to the powerhouse, where it splits into two branches that feed two Francis turbines rated 2 MW and 4 MW. Splitting the capacity lets the plant generate across a wide range of flows. The 2 MW unit runs alone at low flows down to about 120 cubic feet per second (cfs), the 4 MW unit takes over, between 350 and 500 cfs, and both units run above 500 cfs. An underground 25-kV line about 1,200 ft long connects the plant to the local utility station. The developer completed construction in about 11 months [17].

3.1.2.2 Ridgway Dam Hydroelectric Project

The team benchmarked the Ridgway Dam Hydroelectric Project in Colorado, a conversion of an existing Bureau of Reclamation dam on the Uncompahgre River. The dam stores drinking and irrigation water, and the Tri-County Water Conservancy District added an 8 MW hydropower plant to it. The team selected this project because its capacity falls within the competition's 1–10 MW range and it generates power from existing infrastructure. The plant uses two turbine-generator units sized for the river's seasonal flows. An 800-kW unit runs on winter flows of 30 to 60 cfs, and a 7.2 MW unit uses about 500 cfs of summer flow. Sizing one unit for each season lets the plant keep generating year-round instead of shutting down when flow drops. The project cost $18 million [18].

3.1.2.3 James W. Broderick Hydroelectric Project

The team benchmarked the James W. Broderick Hydroelectric Power Facility in Colorado, a conversion of the Bureau of Reclamation's existing Pueblo Dam on the Arkansas River. The Southeastern Colorado Water Conservancy District added a 7.5 MW hydropower plant to the dam, and the plant began operating in 2019. The team selected this project because its capacity falls within the competition's 1–10 MW range and it generates power from water the dam already releases. The plant connects to the dam's North Outlet Works and uses three turbines and two generators that can run individually or together. This setup lets the plant use releases anywhere from 35 to 810 cfs, so it keeps generating as releases change. The plant produces about 28 million kWh per year, and the district financed it with a $17.2 million low-interest loan [19].

3.1.3 Sub-system Benchmarking

3.1.3.1 Lepenci River Francis turbines

The team benchmarked the Francis turbines at the Lepenci River plant in Kosovo because the plant's 10 MW capacity falls within the competition's range. The plant uses three Francis turbines sized for different flows: two with design flows of 3.9 m^3/s and one with a design flow of 1.2 m^3/s, operating at a gross head of 138.6 m. Like the system-level benchmarks, the plant includes a smaller turbine so it can keep generating during the low-flow seasons of the year. The study also compares three orders for connecting and operating the turbines, and it finds that placing the small turbine between the two large turbines uses the most water and produces the most energy per year [20]. This benchmark shows the team that a single turbine cannot capture the full range of a river's flow over a year, so turbine sizing and the order in which units operate both play a key role in a hydropower plant's output.

3.1.3.2 Conowingo East Fish Lift

The team benchmarked the Conowingo East Fish Lift on the Susquehanna River because it has operated successfully since 1991 and its operators publish detailed reports each season. The lift attracts fish with flow at one of several entrance gates. Operators choose the entrance based on which turbines are running, because turbine discharge can disrupt the attraction flow at the other entrances. Inside the lift, a crowder screen moves fish over a hopper, which lifts them into an exit trough. The hopper normally cycles about twice per hour, and operators raise it more often when large numbers of fish arrive. Fish then swim past a counting window, where biologists identify and count each fish by species to report to the resource agencies. Gates downstream of the window control when fish leave the trough, and staff can net invasive species out of the hopper before they reach the reservoir. The report also shows that the lift requires a dedicated three-person crew and that the dam maintains minimum river flows during the fish passage season [21].

3.1.3.3 Bhatghar dam

The Bhatghar Dam in India provides an excellent engineering benchmark for the reliability and longevity of fixed-wheel vertical lift gates. According to Mahajan (2017), the dam utilizes 81 fixed-wheel vertical lift gates to control water discharge through its spillway system, with the gates successfully operating for over 100 years without failure. This service life is attributed to careful engineering design, manufacturing, installation, and consistent maintenance practices. Fixed-wheel vertical lift gates utilize wheels to guide vertical movement and reduce friction between the gate and its supporting structure, allowing for controlled opening and closing under hydraulic loads. The Bhatghar Dam demonstrates that these gates can provide a durable and reliable solution for regulating water flow in large hydraulic structures. For the proposed projects this gate could potentially outlast the dam, though further analysis will have to be done

to ensure closure in emergency conditions

3.1.4 Benchmarking summary

The system-level benchmarks show the team complete, operating NPD conversions in the competition's 1–10 MW range, and the sub-system benchmarks show proven components that make those plants work. Based on the Ridgway and Broderick projects, which cost $18 million and $17.2 million [18], [19], the team sets an estimated budget of $18 million for its conversion. Mahoning Creek finished construction in about 11 months [17], so the team plans for a one-year construction schedule. The developer named permitting as the main source of delay [17], so the team will also research the licensing requirements for its dam to understand their cost and timeline.

Every benchmarked plant uses multiple turbines to keep generating at low flow rates, so the team will research into the feasibility of installing more than one turbine and develop a plan to bring each unit online at a separate flow rate, as Mahoning Creek does [17]. The Lepenci study shows that the order in which turbines operate changes the plant's yearly energy output [20], so the team will study which arrangement gets the most energy from its site's flows. Geotechnical conditions affected the Mahoning Creek penstock [17], so the team will work with a geologist to evaluate the dam's construction and the surrounding ground. If the design includes a fish lift, the team will coordinate with the agency responsible for fish passage to count fish and keep invasive species out of the reservoir, as operators do at Conowingo, and will budget and assign the workers needed to run it [21]. The team uses these benchmarks as a starting point for its own design and builds on them to develop an effective plan for its NPD conversion.

3.2 Literature Review

3.2.1 Zachery Johnson

[2] Short-cut design of small hydroelectric plants

Voros, Kiranoudis, and Maroulis examine how the characteristics of a river and its dam determine the power available at micro and small hydropower plants. The authors consider factors such as turbine efficiency, head, and a flow rate that varies seasonally and from year to year, and they show how designers can use these factors to develop a functional plant. From this analysis, the authors derive short-cut equations that find the optimum design using only data the river itself provides. The team uses one of these equations to calculate the nominal design flow, which accounts for the turbine type and sets the turbine size that delivers the most power for the lowest cost. The team treats this result as a benchmark and refines it as further analysis of each dam provides more site information.

[3] Water Data for the Nation

The U.S. Geological Survey operates this national database, which provides measured streamflow and water-level readings from gauges on rivers across the country. Each gauge records data such as daily discharge and water height, often over decades. The team uses this data to obtain 30-year flow records for each candidate river and builds flow duration curves from them to analyze how much water the power plant can rely on.

[4] Hydroelectric Power Plant

This online course covers siting considerations for hydropower plants, along with the main plant components and how they work together to make a site functional and feasible. The course also shows how to turn river flow data into charts, such as flow duration curves, that guide decisions about a site's design and feasibility. The team uses this course as a guide for building flow duration curves from river data, which assist to size the turbine and shape the plant design based on seasonal changes in flow.

[5] Measurement of Head and Flow

This chapter of the Wyoming Small Hydropower Handbook shows how designers use a flow duration curve to choose an appropriate design flow and turbine for small hydropower sites, noting that a common starting point is the flow equaled or exceeded about 30% of the time. It also explains how to measure head and how friction losses reduce the gross head to the net head that the turbine must be designed for. The team uses this reference as secondary support to validate its site selection and its penstock and turbine designs.

[6] A closer look at run-of-river hydro schemes

Faircloth of Dulas discusses how designers plan run-of-river hydropower plants around the natural flow of the river. The author explains how seasonal changes in flow affect a plant's practical operation and outlines general guidelines for sizing turbines, noting that most schemes are designed around the flow exceeded 35–40% of the time (Q35–Q40) for the best economic return. The team uses this source to validate its selected design flow and to show that most small hydropower plants do not run at full capacity year-round, operating at part load during lower flows.

[7] Guide on How to Develop a Small Hydropower Plant

This guide from the European Small Hydropower Association walks through the full development of a small hydropower plant. It covers hydrology, turbine types and selection, specific speed, efficiency, and the minimum operating flow of each turbine type, and it also addresses the environmental factors developers must consider. The team uses this guide as one of the main sources for turbine design and for consideration as it applies in final site selection and plant design.

[8] Design models for small run-of-river hydropower plants: a review

This review examines the design models that engineers use for small run-of-river hydropower plants, drawing on previous studies and existing plants as references. It covers the site characteristics and plant characteristics a successful design must account for, including turbine head ranges, minimum operating flows, and methods for choosing the design discharge. The team uses this review as an additional main source in designing the hydropower plant and to cross-check the design process against the other sources.

[9] Hydropower Baseline Cost Modeling, Version 2

Oak Ridge National Laboratory presents empirical equations for the initial capital cost and the annual operation and maintenance cost of U.S. hydropower projects, organized by project type, with a specific set of equations for non-powered dam conversions. The equations estimate costs from a plant's installed capacity and head, giving preliminary estimates for early project planning. The team uses these equations to build its construction budget and its cost-of-energy analysis.

[9] Small Hydropower cost Reference Model

Oak Ridge National Laboratory provides cost equations for turbine-generator packages by turbine type, based on manufacturer budget prices for hundreds of projects, along with typical cost breakdowns for projects at existing sites. The team uses these equations to price the turbine-generator package and uses the cost breakdowns to divide the total construction cost into the Bill of Materials categories.

[10] Hydropower Licensing Timeline and Cost Dataset

Oak Ridge National Laboratory compiles licensing data for over 100 U.S. hydropower projects, including how long each project took to move through the Federal Energy Regulatory Commission (FERC) licensing process and the reported costs of licensing and the required regulatory studies. The team selected the original licenses for non-powered dam projects between 1 and 10 MW, the size range of its design, and averaged their costs and timelines to build the licensing portion of the project budget.

[11] Hydro Projects Get US $663,881 from Massachusetts

The article reports state funding for small hydropower feasibility studies in Massachusetts, including the Pittsfield ($44,000) and Fitchburg ($15,600) studies. The team uses these as real-world examples of siting study costs.

[12] Dam shows potential for green energy

Horn describes a $30,000 feasibility study of hydropower at Taylor Park Dam, Colorado. The team uses this study as another real-world data point for the cost of evaluating an existing dam for hydropower.

[13] Gouverneur Hydroelectric Upgrade and Expansion Feasibility Study

The New York Regional Economic Development Councils describe a $25,000 award for a study of upgrading and expanding a village's existing hydropower system. Together with [11] and [12], this study forms the basis for the siting cost for the team.

[14] Simple Levelized Cost of Energy (LCOE) Calculator Documentation

The National Laboratory of the Rockies explains a simple method for calculating the levelized cost of energy, which is the cost to produce each unit of electricity over a plant's life. The method spreads the construction cost into equal yearly payments, adds the yearly operation and maintenance cost, and divides by the energy the plant produces each year. The team applies this method across a range of trial design flows at each candidate site and selects the design flow with the lowest cost of energy as the best turbine size.

[15] Performance of fish passage structures at upstream barriers to migration

Bunt, Castro-Santos, and Haro perform a review of 19 fish passage monitoring studies comparing how well fish lift, vertical-slot, and nature-like fishways attract fish and pass them upstream. The team uses the performance data to compare fish passage options.

[16] Upstream Fish Passage Technologies for Managed Species

The Atlantic States Marine Fisheries Commission's Fish Passage Working Group reviews upstream fish passage technologies, including fish ladders, nature-like fishways, and fish lifts, and compares their advantages and limitations. The report considers how well different fish species can swim through each type of passage, and it notes that fish lifts also allow operators to sort and count fish by species. The report also notes that lifts are generally preferred at dams over about 25 ft, and the team uses this guideline to select a fish lift for the dam.

[17] Adding Power to a Non-Powered Dam: Mahoning Creek

Johnson, Garner, Collins, and Strough describe how Enduring Hydro converted the U.S. Army Corps of Engineers' Mahoning Creek flood control dam in Pennsylvania into a 6 MW hydropower plant. The authors cover the turbines, penstock, and power line that make up the plant, along with the design choices behind them. They also explain the challenges the developer faced during permitting and construction. The team uses this source as its primary system-level benchmark because it covers the project's design, schedule, and results in depth.

[18] Tri-County Water Commissions 8-MW Ridgway Dam Small Hydropower Plant

Harris reports on the Tri-County Water Conservancy District's 8 MW hydropower plant at the Bureau of Reclamation's Ridgway Dam in Colorado. The article describes how the plant uses one small unit for winter flows and one large unit for summer flows, and it covers the project's $18 million cost. The team uses this source as a system-level benchmark that shows how sizing units to seasonal flows keeps a plant generating year-round, and as a reference for the expected cost of a conversion in the competition's size range.

[19] 7.5-MW James W. Broderick Hydroelectric Power Facility Now Operating in Colorado

This article reports on the Southeastern Colorado Water Conservancy District's 7.5 MW plant at the Bureau of Reclamation's Pueblo Dam. It explains how the plant uses three turbines and two generators to cover a wide range of flow rates. The team uses this source as a system benchmark for a recent conversion that generates power from water the dam already releases.

[20] The optimal order configuration of turbines in a small hydropower plant: A case study of the Lepenci River in Kosovo

Bresa, Dragusha, Berisha, and Selimaj study a 10 MW run-of-river plant in Kosovo that uses three Francis turbines of different sizes, and they compare the order in which the turbines operate to find the arrangement that produces the most energy. The team uses this source as its sub-system benchmark for Francis turbines.

[21] Summary of Operations at the Conowingo Dam East Fish Passage Facility, Spring 2020

Normandeau Associates reports on the 2020 season of the Conowingo Dam East Fish Lift for Exelon Generation. The report describes how the lift attracts fish, moves them through the crowder and hopper, and counts each fish by species at a viewing window before release, and it notes the invasive species concerns. The team uses this source as its sub-system benchmark for fish lifts.

3.2.2 Ben Krotonsky

[22] An Assessment of Technical Hydropower Potential at Non-Powered Dams in the United States

Oak Ridge National Laboratory performed a mass analysis of ~2700 non-powered dams (NPD) in the United States, determining each one's potential to be converted into a hydropower plant. The dataset includes valuable insight into dam conditions, such as stream flow statistics at the site, expected power output, expected capacity, and nominal flow. These conditions are extremely valuable to the project, as it provides an all-encompassing repository containing not only viable sites, but also statistics that we can use to further narrow the list of dams to fit the project scope.

[23]NPD Hydro: A toolkit for pre-feasibility evaluation of non-powered dams

This interactive tool, developed by the Idaho National Laboratory and the Pacific Northwest National Laboratory in partnership with the DOE, gives extensive information relevant to NPD conversion, past what the ORNL database could provide. This tool goes a step further, giving insights into environmental impact and licensure. In addition, the methodology used for technical site analysis provides a framework that will elevate the quality of our production estimations.

[24] OpenStreetMap Power Substation Data

This source provides a database of electrical substation locations across the U.S., excluding minor distribution, traction, industrial, and converter substations. This allows us to gauge a viable substation for a given dam selection and determine its approximate distance from the site.

[25] National Renewable Energy Laboratory utility energy rates

This tool gives reference to energy cost in the area surrounding the dam, based on utility energy rates in the ZIP code of the site. Using this cost, we can determine energy needs in the immediate community, as higher rates imply either low supply or high demand.

[26] Turbine selection chart

The website provides a figure that shows the optimal turbine given a stream's head and flow. We can use the chart to qualitatively determine the optimal turbine for a dam, based on its stream flow data.

[27] U.S. Geological Survey, Protected Areas Database of the United States (PAD-US)

The U.S. Geological Survey's database shows protected and restricted land boundaries within the United States. It covers designations that would pose significant difficulties to dam retrofitting, such as national parks, wilderness areas, wild and scenic rivers, wildlife refuges, tribal land, and state parks. We can use these boundaries to disregard dams located inside a protected area. This filter eliminates sites with significant legal barriers and reduces environmental risk.

[28] National Inventory of Dams (NID)

This U.S. Army Corps of Engineers database catalogs dams across the nation, including both hydropower and non-powered dams (NPDs). Each record lists the dam's NID ID, name, state, waterway, and coordinates, along with details such as ownership and construction type. The NID ID is the key that links our NPD list to the ORNL dataset, and the waterway, state, and coordinates let us locate and identify each site. We can use this database to cross-check our NPD list and explore aspects of site selection that the ORNL analysis did not cover.

[29] Little Falls Dam fish passage system

The Little Falls Dam has an existing fish passage system, which was restored and completed in 2014. This source provides insight into current conditions of the dam and proves that the conversion would not need new implementation of a fish passage.

[30] Darcy-Weisbach equation pipe roughness estimation

The Darcy-Weisbach equation is a common method for head loss estimation, determining frictional head loss for a flow through a pipe of a given material. I used the equation and roughness estimation to determine the expected head loss through the penstock.

3.2.3 Tyler Loggins

[31] 2027 Hydropower Collegiate Competition

The U.S. Department of Energy (DOE) provides the official guidelines and requirements for the 2027 Hydropower Collegiate Competition. These guidelines explain the Siting, Design, and Community Connections Challenges, along with the required deliverables. We use this information to understand the competition requirements, establish project goals, and ensure our site selection and design meet the competition's expectations.

[32] Technical Potential for Hydropower Capacity at Nonpowered Dams

Oak Ridge National Laboratory (ORNL) provides a dataset evaluating the potential for hydropower generation at existing non-powered dams throughout the United States. The dataset includes information on hydraulic head, flow rate, and estimated generation capacity. We can use this information to compare potential sites, verify our preliminary power calculations, and narrow our selection to dams that meet the project's requirements.

[33] Small/Low-Impact Hydropower Projects

The Federal Energy Regulatory Commission (FERC) provides information on the regulations and licensing processes for developing small hydropower facilities. This includes environmental requirements, licensing exemptions, and other considerations involved in converting existing dams. We can use this information to identify regulatory restrictions and evaluate how they could affect the feasibility of our selected site.

[34] U.S. Bureau of Reclamation Hydropower Program

The Bureau of Reclamation provides information on the operation, maintenance, and development of existing hydropower facilities. This source explains how water infrastructure is used for electricity generation and discusses considerations for maintaining efficient operation. We can use this information to better understand existing hydropower systems and apply established practices to our preliminary design.

[35] Performance Testing Multiple Units of Similar Hydraulic Design

This Bureau of Reclamation report examines methods used to evaluate the performance of hydraulic turbines with similar designs. It discusses how performance data can be used to assess turbine efficiency and identify differences between operating conditions. We can use these methods to support our performance analysis and compare results from our mathematical models and fluid simulations.

[36] Fluid Mechanics — White and Xue

This textbook explains the fluid mechanics principles used to analyze flow through pipes and hydraulic systems. It covers Bernoulli's equation, conservation of mass, pipe friction, and hydraulic head losses. I used these concepts to develop our preliminary penstock calculations, including cross-sectional area, water velocity, and frictional head loss using the Darcy-Weisbach equation. These calculations help

determine the available net head and expected power generation at potential dam sites.

[37] Environmental Design of Low-Head Run-of-River Hydropower in the United States

This research examines the environmental factors that must be considered when developing low-head hydropower facilities. It discusses fish passage, sediment transport, water quality, and the effects of hydropower development on surrounding ecosystems. We can use this information to evaluate environmental risks at potential dam sites and develop design concepts that reduce environmental impacts while maintaining energy production.

3.2.4 Zachary Mills

[46] ORNL National Hydropower Fish Passage Database

Oak Ridge National Laboratory (ORNL) provides a national database of fish passage facilities at existing hydropower developments throughout the United States. The database includes information on passage facility types, target fish species, operating schedules, and facility costs. We can use this information to investigate fish passage infrastructure at our selected non-powered dams and determine whether existing facilities may affect the environmental feasibility of each site.

[47] National Hydropower Fish Passage Database User's Guide and Methodology

Oak Ridge National Laboratory explains how the National Hydropower Fish Passage Database was developed, organized, and maintained. The guide describes how fish passage information is collected from different sources and matched to individual hydropower facilities. We can use this information to interpret the database correctly, identify existing fish passage measures, and understand the limitations of the available data when evaluating our selected sites.

[48] How Do Fish Survive Hydropower Dams?

The U.S. Department of Energy discusses the challenges fish face when moving through and around hydropower facilities and describes technologies developed to improve fish passage. These technologies are intended to reduce the environmental impacts of hydropower while maintaining electricity generation. We can use this information to evaluate potential environmental concerns associated with turbine selection and identify design approaches that could improve fish survival at our selected sites.

[49] A Census of Fish Passage Facilities at U.S. Hydropower Developments

This research examines the availability and distribution of fish passage facilities at hydropower developments across the contiguous United States. The study identifies the types of facilities used to support fish migration and examines patterns in their implementation. We can use these findings to compare fish passage infrastructure across existing hydropower developments and determine what types of environmental mitigation measures may be relevant to our proposed dam retrofits.

[50] Improving Fish Migration at Hydropower Dams

NOAA Fisheries explains how fish migration can be improved at hydropower dams through fish passage structures, improved facility operations, and coordination between regulatory agencies and hydropower operators. The source discusses the importance of maintaining access to upstream and downstream habitats while continuing to generate electricity. We can use this information to identify potential mitigation strategies and consider how fish passage requirements could influence the design and operation of our proposed hydropower systems.

[51] Barriers to Fish Migration

NOAA Fisheries describes how dams and other infrastructure can prevent fish from reaching the habitats necessary for reproduction and survival. The source explains the effects of migration barriers on fish

populations and discusses approaches such as fish passage improvements and barrier removal. We can use this information to evaluate the environmental concerns associated with our candidate sites and identify potential restrictions that should be considered before recommending a dam for hydropower development.

[52] FishSafe Turbine Design

Natel Energy describes its FishSafe turbine technology, which is designed to allow downstream fish passage through the turbine while maintaining hydraulic performance. The company discusses turbine design features intended to reduce fish injury and improve passage survival. We can use this information to investigate fish-friendly turbine technology and compare its potential benefits with conventional turbine designs. This information may also help identify design alternatives that balance electricity generation, turbine efficiency, and environmental considerations.

3.2.5 Robert Wanek

[38] 100 Successful Years of Vertical Lift Gates of Bhatghar Dam—Design, Manufacturing, Erection, and Maintenance: A Case Study

This article examines the design, construction, and maintenance of the fixed-wheel vertical lift gates used at Bhatghar Dam in India, highlighting their successful operation for over 100 years. It demonstrates the long-term durability and reliability of fixed-wheel vertical lift gates, making them a valuable benchmark for evaluating water-control systems in hydroelectric dam applications.

[39] Dams and Tribal Land Loss in the United States

This article examines the social and environmental impacts of dam construction on Indigenous communities in the United States, identifying over 1.13 million acres of tribal land submerged by reservoirs associated with 424 dams. It highlights the importance of considering Indigenous land rights, environmental justice, and community impacts when evaluating existing dams and planning future hydroelectric infrastructure projects.

[40] Mechanical and Electrical Design for Civil Works Structures – U.S. Army Corps of Engineers (EM 1110-2-2610)

This engineering manual provides guidelines for designing and operating mechanical and electrical systems used in dams, navigation locks, and other hydraulic structures, including vertical lift gates and their operating machinery. It discusses gate operating mechanisms, hydraulic loading conditions, and safety considerations, making it a valuable reference for evaluating gate systems capable of controlling or completely stopping water flow in hydroelectric applications.

[41] Global Hydropower Expansion Without Building New Dams

This article examines the potential to increase global hydroelectric power generation by retrofitting existing non-powered dams and improving the efficiency of current hydropower facilities, estimating a possible 9% increase in global hydropower capacity. It emphasizes the environmental and economic benefits of upgrading existing infrastructure rather than constructing new dams, reducing construction costs while minimizing impacts on river ecosystems and biodiversity.

[42] Impact of Installation Deviations on the Dynamic Characteristics of the Shaft System for 1 Gigawatt Hydro-Generator Unit

This article investigates how installation errors and misalignment affect the vibration, stability, and dynamic performance of large hydroelectric generator shaft systems. It highlights the importance of precise shaft alignment and installation procedures to minimize mechanical stress, prevent equipment damage, and ensure reliable long-term operation of hydroelectric power generation systems.

[43] Hydroelectric Power Plants: Mechanical Design – U.S. Army Corps of Engineers (EM 1110-2-4205)

This engineering manual provides design guidelines and selection criteria for mechanical equipment used in hydroelectric power plants, including Francis and Kaplan turbines, generators, governors, and penstock

shutoff valves. It emphasizes equipment reliability, operational safety, and proper mechanical system design, making it a valuable reference for evaluating turbine configurations, shaft systems, and water-flow control mechanisms in hydroelectric dam applications.

[44] Shigley's Mechanical Engineering Design, 10th Edition – Richard G. Budynas and J. Keith Nisbett

This textbook provides fundamental engineering principles, equations, and design methods for mechanical components, including shafts, gears, bearings, fasteners, and power transmission systems. It serves as a valuable reference for calculating torque, shear stress, shaft diameters, and factors of safety, supporting the mechanical design and structural evaluation of hydroelectric turbine and generator systems.

[45] Free House of Quality Templates (Excel, PDF, and PowerPoint) – Shane Drumm, ProjectManagers.net

This article provides an overview of the House of Quality (HOQ) methodology, explaining how customer requirements are translated into measurable engineering specifications through Quality Function Deployment (QFD). It offers templates and evaluation techniques for prioritizing design requirements, comparing engineering concepts, and identifying technical trade-offs, making it a useful resource for evaluating hydroelectric dam components and selecting optimal design solutions.

3.3 Mathematical Modeling

3.3.1 Flow Duration Chart and Turbine Design Flow for Cost Analysis (Zach J)

3.3.1.1 Tools

The team pulled 30 years of annual mean flow data for each of the three top candidate sites from USGS gauge records [3]. Using the flow duration method taught in the Udemy course [4], the team built flow duration curves in MATLAB to show how often each river equals or exceeds a given flow. The team then wrote a second MATLAB code that calculates the annual energy output across a range of design flows. For each design flow, the code calculates the construction and operation and maintenance (O&M) costs with ORNL's NPD cost equations [9] and the levelized cost of energy [15]. The code then plots annual energy and cost of energy against design flow, which shows the turbine size that produces energy at the lowest cost. The team compared these results against two baseline designs: the short-cut nominal design flow and the flow exceeded 30% of the time (Q_{30}).

3.3.1.2 Nominal design flow

$$Q_{nom} = \left[\frac{\gamma q_{50}}{(1 + (\gamma - 1)q_{50})(1 - q_{min})} + q_{min}\right] Q_{max} \quad (1)$$

The team calculated the nominal design flow with the short-cut method [2]. The method uses the river's median, minimum, and maximum flows, together with a coefficient for the chosen turbine type, to estimate a cost-effective turbine size. The team used this design flow as a baseline in the charts in Figure 4. Each term in the equation is defined below.

$$q_{50} = \frac{Q_{50}}{Q_{max}} \quad (2)$$

q_{50} is the ratio of the median flow to the maximum flow over the period of record.

$$q_{min} = \frac{Q_{min}}{Q_{max}} \quad (3)$$

q_{min} is the ratio of the minimum flow to the maximum flow over the period of record.

γ is a coefficient assigned to each turbine type: 0.422 for Francis, 0.364 for Kaplan, and 0.369 for Pelton turbines [2].

3.3.1.3 Flow Duration Curve

$$P_{e(Percent\ of\ exceedence)} = \frac{m}{n+1} * 100 \quad (4)$$

This equation gives the percentage of time the river equals or exceeds a given flow [4]. The team ranked the annual mean flows from highest to lowest, where m is the rank of each flow and n is the number of years of record. Plotting each flow against its exceedance percentage produces the flow duration curve shown in Figure 3.

3.3.1.4 power output

$$P(kW) = Q(cfs) * H(ft) * \frac{\eta}{11.8} \quad (5)$$

The team calculated the installed power capacity with the hydropower equation converted to U.S. units [6]. In this equation, P is the power in kilowatts, Q is the flow in cubic feet per second, H is the head in feet, and η is the overall plant efficiency, assumed to be 85%.

3.3.1.5 Cost and design Flow Verification

The team estimated the initial capital cost and the annual O&M cost with ORNL's equations for non-powered dam conversions [8], shown below. Both equations give costs in 2014 dollars, so the team escalated the results to 2026 dollars at an assumed 2% per year.

$$Initial\ Capital\ Cost\ (ICC) = 11{,}489{,}245 \cdot P(MW)^{0.976} \cdot H(ft)^{-0.240} \quad (6)$$

$$Operation\ and\ Maintenance(O\&M) = 225{,}417 \cdot P(MW)^{0.547} \quad (7)$$

To find the cost of energy, the team spread the capital cost over an assumed 30-year plant life at a 7% discount rate using the capital recovery factor (CRF), then applied the levelized cost of energy equation [14]:

$$Capital\ Recovery\ Factor(CRF) = \frac{i(1+i)^n}{(1+i)^n - 1}, i = 7\%, n = 30\ years \quad (8)$$

$$Levelized\ Cost\ of\ Energy\ (LCOE) = \frac{(ICC \cdot CRF + O\&M)}{E}; E = (annual\ energy\ produced\ by\ turbine) \quad (9)$$

In these equations, i is the discount rate, n is the plant life in years, and E is the annual energy the plant produces.

3.3.1.6 Flow Duration and Cost and Design Charts

Figure 3 shows the flow duration curve for New Croton Dam, one of the team's candidate sites, built from 30 years of USGS annual mean flow data [3]. The team will use this curve to select specific turbine designs once it finalizes site selection. The horizontal axis shows the percentage of time the river equals or exceeds a given flow, and the vertical axis shows the annual mean flow in cubic feet per second. The red dashed line marks the nominal design flow from the short-cut method [2], and the green line marks the flow exceeded 30% of the time (Q_{30}), a common industry rule of thumb for design flow [5]. The dotted lines below each design flow mark the turbine's minimum operating flow, below which the turbine must shut down [7][8]. The boxes on the right show the percentage of years a turbine sized for each design flow would run at full power, at part load, or shut down. The team includes the nominal and Q_{30} design flows only as benchmarks. The team selects the final design flow from the cost-of-energy analysis in Figure 4, and the possible use of multiple turbines will also shape the flows each unit is designed for. The flow duration curves for Little Falls Dam and Birch Hill Dam appear in Appendix B (Figures 29 and 31).

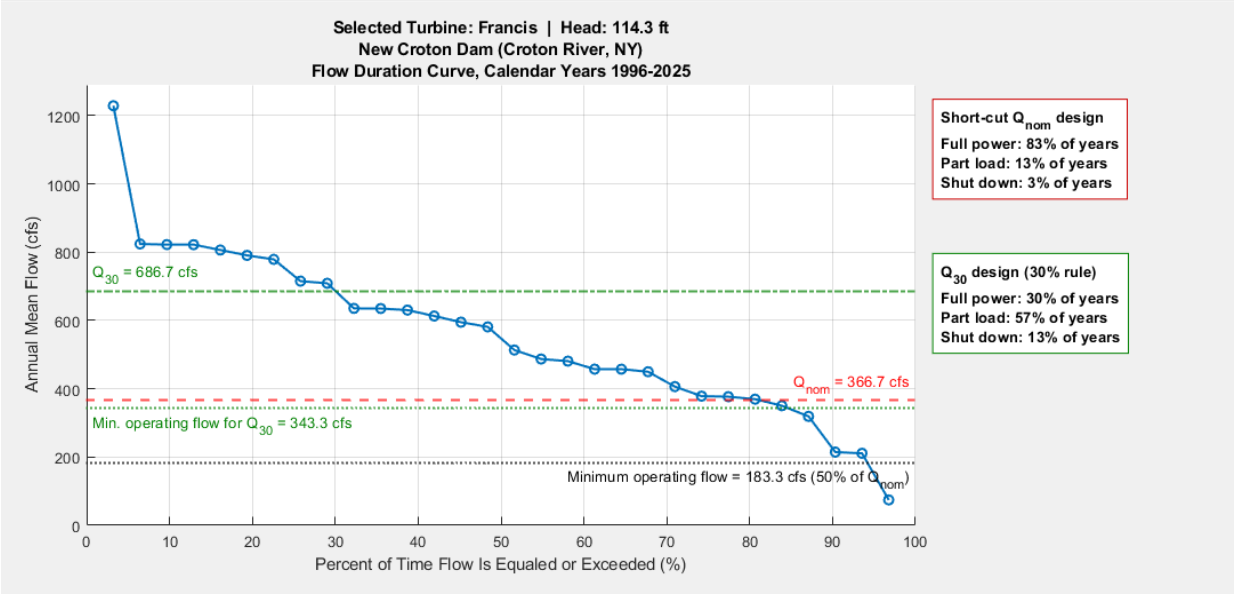


Figure 3: New Croton Dam Flow Duration chart

Figure 3 shows the design flow analysis the team produced in MATLAB for New Croton Dam. The horizontal axis shows the trial design flow of the turbine in cubic feet per second. The blue line, read on the left axis, shows the annual energy the turbine would generate at each design flow in GWh per year. The orange line, read on the right axis, shows the levelized cost of energy in dollars per kWh, calculated from ORNL's NPD cost equations [9] and the LCOE method [15]. The vertical lines mark three design flows: the red dashed line shows the short-cut nominal flow [2], the green dash-dot line shows the Q_{30} flow [5], and the black dotted line shows the design flow with the lowest cost of energy. The cost curve first falls as larger turbines capture more of the river's flow, then rises once the added turbine size costs more than the extra energy it produces. The lowest point on this curve sets the most cost-effective turbine size for the site. The design flow analyses for Little Falls Dam and Birch Hill Dam appear in Appendix B (Figures 30 and 32).

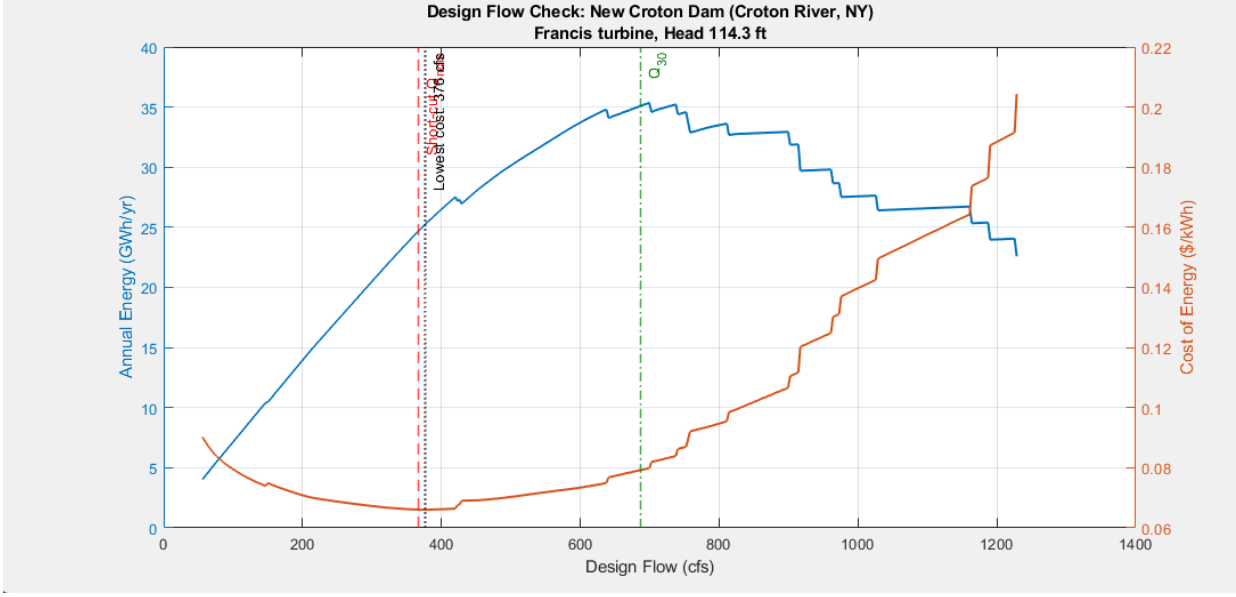


Figure 4: New Croton Dam -Power Output and Energy Cost at Specific Design Flows

3.3.1.7 Mathematical Modeling Future work

The team will extend the model with daily mean flow data from the USGS gauge at each site [3]. It will build flow duration curves for each year and each season to show how flow changes through the year, including low-flow periods that annual means hide. Together with the energy and cost-of-energy analysis, these seasonal curves will show whether the plant needs multiple turbines to keep generating through the low-flow seasons. The team will model scenarios ranging from a single turbine to several turbines of different sizes, and it will compare each scenario's annual energy output against its cost to find the most cost-effective arrangement.

The team will contact the owner of each candidate dam to request measured upstream and downstream water levels. The current model uses head values from the site-ranking dataset, so measured water levels will let the team calculate the actual head at the site and how it changes through the year. The team will also interview industry professionals and adjust the model's assumed plant life, discount rate, and cost uncertainty based on their input. These updates will make the power output, construction cost, and cost-of-energy results more accurate before the team finalizes the turbine design.

3.3.2 Overall System Efficiency Estimation (Ben)

In hydropower generation, there are different efficiencies which play into the overall power output of the system. The energy conversion efficiency, from when the flow originally enters the penstock to when the energy meets the grid, experiences energy loss at every stage:

1. Head loss through the penstock
2. Losses through the turbine
3. Losses through the generator

Each can be quantified, in their own respect, to determine how much of the flow is being converted into usable electricity for the grid. The analysis is completed for the New Croton Reservoir Dam in New York.

3.3.2.1 Assumptions

The following assumptions cover all four phases of the analysis. Additionally, we assume that the connection between the turbine and the generator is direct-drive, meaning there is no gear box, the flow rate and head are constant, and the flow is incompressible and steady.

Table 1: Assumptions for overall efficiency analysis

Variable	Value
Volumetric Flow Rate	$Q = Q_{30} = 11.48\, m^3/s$, ORNL database
Gross Head	$H_g = 34.84\, m$, ORNL database
Gravitational constant	$g = 9.81\, m/s^2$
Water density	$\rho = 1000\, kg/m^3$ (STP)
Kinematic viscosity	$\nu = 1.002 \times 10^{-6}\, m^2/s$ (STP)
Penstock Length	$L = 50m$
Penstock Diameter	$D = 2m$
Penstock Roughness	$\epsilon = 0.15mm$, concrete
Fitting Loss Coefficient	$K_{elb} = 0.2$, 45° elbow
Entrance Loss Coefficient	$K_{ent} = 0.5$, squared entrance
Target (Peak) Efficiency Factor	$\eta_{turb} = 94\%$, Francis turbine (Fig. 34)
Specific Speed at Peak	$\Omega_{peak} = 1.4\, rad/s$
Grid Frequency	$f_{grid} = 60\, Hz$, US Grid

3.3.2.2 Penstock Efficiency Factor

From the assumptions, we determined the area of penstock, along with the velocity of the fluid.

$$A = \frac{\pi D^2}{4} = \frac{\pi \cdot 2^2}{4} = 3.14\, m^2 \tag{10}$$

$$V = \frac{Q}{A} = \frac{11.48}{3.14} = 3.65\, m/s \tag{11}$$

To determine whether the flow is turbulent or laminar, we calculated the Reynolds number of the water.

$$Re = \frac{VD}{\nu} = \frac{3.65 \cdot 2}{1.002 \times 10^{-6}} = 7.29 \times 10^6 \tag{12}$$

$$\because Re > 4000 \therefore Turbulent$$

Referring to the Moody chart in Figure 33, coupled with a relative roughness of $\epsilon/D = 0.75 \times 10^{-3}$, the Darcy frictional coefficient is estimated at about $f = 0.0125$. In addition, assuming a penstock containing a squared entrance and two 45° elbow fittings, the loss coefficient sum for the system is expected to be around $\sum K = 0.9$. These values can be applied to the head loss equation, where:

$$h_L = \left(f\frac{L}{D} + \sum K\right)\frac{V^2}{2g} = \left(0.0125\frac{50}{2} + 0.9\right)\frac{3.65^2}{2 \cdot 9.81} = 0.823\, m \tag{13}$$

Applied to determine the efficiency factor for penstock:

$$H_{net} = H_g - h_L = 34.84 - 0.823 = 34.017\, m \tag{14}$$

$$\eta_{pen} = \frac{H_{net}}{H_g} = \frac{34.017}{34.84} = 0.976 \tag{15}$$

3.3.2.3 Generator Efficiency Factor

Assuming the use of a Francis turbine, the optimal specific speed at peak efficiency (94%) is about 1.4 rad. This can be used to determine the necessary pole count for the generator, provided a direct-driven connection between it and the turbine. First, we evaluated for the peak rotational speed:

$$N_{peak} = \frac{60\Omega_{peak}(gH_{net})^{3/4}}{2\pi\sqrt{Q}} = \frac{60 \cdot 1.4 \cdot (9.81 \cdot 34.017)^{3/4}}{2\pi\sqrt{11.48}} = 308.1\, rpm \tag{16}$$

We then used this peak rotational speed to determine the optimal pole count to achieve this:

$$p = \text{round}\left(\frac{120f}{N_{peak}}\right) = \text{round}\left(\frac{120 \cdot 60}{308.1}\right) = 23.4\, poles \approx 24\, poles \tag{17}$$

Generators with similar operating capacities reported a 99% efficiency. Thus, $\eta_{gen} = 0.99$

3.3.2.4 Overall Efficiency

We determined the overall efficiency by multiplying all the efficiency factors together and estimated real power output by multiplying this total factor by the capacity listed in the ORNL database.

$$\eta_{total} = \eta_{pen} \cdot \eta_{turb} \cdot \eta_{gen} = 0.976 \cdot 0.94 \cdot 0.99 = 0.908 = \eta_{total} \tag{18}$$

$$P_{real} = \eta P_{listed} = 0.908 \cdot 5.06 = 4.60\, MW = P_{real} \tag{19}$$

3.3.3 Turbine Selection and Preliminary Site Analysis (Zach M)

3.3.3.1 Tools and Selection Criteria

The purpose of this analysis is to evaluate the hydraulic characteristics of the three highest ranked hydropower sites, New Croton Reservoir Dam, Little Falls Dam, and Birch Hill Dam, and identify suitable turbine types for further investigation. The analysis considers hydraulic power, average flow, available head, and the variability of flow and head. These factors provide a preliminary basis for comparing turbine options and identifying the operating characteristics that should be considered during the design process.

The site data were obtained from the team's hydropower site dataset. The calculations will be performed in MATLAB using the same method for all candidate sites to maintain consistency in the comparison.

3.3.3.2 Hydraulic Power Calculations

The theoretical hydraulic power available at a site is calculated using:

$$P_h = \frac{\gamma Q H}{550} \tag{20}$$

where P_h is theoretical hydraulic power in horsepower hp, gamma is the specific weight of water in lbf/ft^3, Q is flow rate in ft^3/s, H is hydraulic head in feet, and 550 is the conversion factor from foot-pounds-force per second to horsepower. For this calculation, the specific weight of water is assumed to be 62.4 lbf/ft^3

New Croton Reservoir Dam is used as an example to demonstrate the calculation. The site dataset reports an average flow of 405.27 ft^3/s and an average head of 114.25 ft. Substituting these values into the hydraulic power equation gives:

$$P_h = \frac{(62.4)(405.27)(114.25)}{550} \tag{21}$$

$$= 5257hp$$

This result can also be expressed in megawatts for comparison with the site's reported installed capacity:

$$P_h(\mathrm{MW}) = 3.92\ \mathrm{MW}$$

This result represents theoretical hydraulic power based on the reported average flow and head, before turbine efficiency and other system losses. The dataset separately reports an installed capacity of 5.058 MW and annual generation of 20,192.47 MWh/yr. These reported values describe the existing site's generation characteristics and are not interchangeable with the calculated theoretical hydraulic power.

The same equation will be applied to Little Falls Dam and Birch Hill Dam using their corresponding average flow and head values. Comparing the theoretical hydraulic power of the sites provides a consistent way to evaluate their hydraulic potential, but it does not independently determine the best turbine or predict actual electrical output.

3.3.3.3 Flow and Head Variability

Average flow and head do not fully describe the operating conditions expected for the design turbine. Flow and head variability are evaluated using their coefficients of variation:

$$CV_Q = \frac{\sigma_Q}{Q_{bar}} \tag{22}$$

$$CV_H = \frac{\sigma_H}{H_{bar}} \tag{23}$$

where sigma Q and sigma H are the standard deviations of flow and head, and Q bar and H bar are their respective mean values. The coefficient of variation measures variability relative to the mean, allowing sites with different average conditions to be compared.

For New Croton Reservoir Dam, the dataset reports a flow coefficient of variation of 0.635972 and a head coefficient of variation of 0.036655. The corresponding standard deviations can be estimated using the reported average flow and head:

$$\sigma_Q = CV_Q Q_{bar} = 257.8 \frac{ft^3}{s} \tag{24}$$

For head:

$$\sigma_H = CV_H H_{bar} = 4.19ft \tag{25}$$

These results indicate that flow is substantially more variable relative to its mean than head at New Croton Reservoir Dam. This matters for turbine selection because a turbine should be capable of operating over the site's expected range of flow and head conditions, rather than being selected solely from average values.

The coefficients of variation will be used to compare the relative stability of flow and head at New Croton Reservoir Dam, Little Falls Dam, and Birch Hill Dam. They will be considered alongside average flow, head, and theoretical hydraulic power rather than used as independent measures of site suitability.

3.3.3.4 Preliminary Turbine Selection

Turbine selection depends on the site's available head, flow rate, and expected operating range. Kaplan turbines are generally suited to lower head, higher flow applications and can accommodate changes in flow through adjustable blade configurations. Francis turbines are commonly used in medium head applications and cover a broad range of operating conditions. Pelton turbines are generally suited to high head, lower flow applications.

New Croton Reservoir Dam has a reported average head of approximately 114.25 ft and an average flow of approximately 405.27ft^3/s. Based on these preliminary hydraulic characteristics, a Francis turbine is a candidate for further investigation. This is an initial engineering assessment rather than a final recommendation. The site's flow variability and the turbine's operating range must be evaluated before a turbine type can be selected confidently.

3.3.4 Penstock Wall Thickness (Robert)

The purpose of these calculations was to determine the preliminary minimum wall thickness required for the penstocks at the Fish Barrier, New Croton, and Cochiti Dam locations. The process involved calculating the penstock's cross-sectional area, water velocity, internal pressure, allowable hoop stress, and minimum wall thickness. The Fish Barrier location was used as the example, with an internal diameter of 2.7 m, a flow rate of approximately 15.57 m³/s, and a steel yield strength of 205 MPa.

3.3.4.1 Cross-Sectional Area and Water Velocity

The cross-sectional area was calculated using the internal penstock diameter. This area was then used to determine the average water velocity through the pipe.

$$A = \frac{\pi D^2}{4} = \frac{\pi(2.7)^2}{4} = 5.726 \text{ m}^2 \tag{26}$$

$$v = \frac{Q}{A} = \frac{15.57}{5.726} = 2.719 \text{ m/s} \tag{27}$$

3.3.4.2 Internal Design Pressure

The internal pressure was calculated using the hydrostatic pressure equation, accounting for water density, gravitational acceleration, and hydraulic head. Surge head was assumed to be zero for these preliminary calculations.

$$[P = \rho g(H + H_s)] \tag{28}$$

$$[P = (1000)(9.81)(18.23 + 0) \approx 0.1788 \text{ MPa}]$$

3.3.4.3 Allowable Hoop Stress

The allowable hoop stress was determined by dividing the material's yield strength by the selected factor of safety of 1.2.

$$\left[S = \frac{S_y}{n} = \frac{205}{1.2} = 170.83 \text{ MPa}\right] \tag{29}$$

3.3.4.4 Minimum Wall Thickness and Stress Verification

Lamé's thick-walled cylinder equation was used to calculate the theoretical minimum wall thickness required to withstand the internal pressure. The resulting thickness was verified using the thin-wall hoop stress equation.

$$\left[t_p = \frac{D}{2}\left[\sqrt{\frac{S+P}{S-P}} - 1\right]\right] \tag{30}$$

$$\left[t_p = \frac{2.7}{2}\left[\sqrt{\frac{170.83 + 0.1788}{170.83 - 0.1788}} - 1\right] \approx 1.414 \text{ mm}\right]$$

$$\left[\sigma_h = \frac{PD}{2t_p} \approx 170.7 \text{ MPa}\right] \tag{31}$$

The calculated hoop stress was approximately equal to the allowable stress of 170.83 MPa, confirming consistency with the preliminary design assumptions.

3.3.4.5 Nominal Wall Thickness and Results

The final nominal thickness equation accounts for corrosion allowance (c) and manufacturing tolerance (m). Both were assumed to be zero in this initial evaluation.

$$\left[t_n = \frac{t_p + c}{1 - m} = \frac{1.414 + 0}{1 - 0} = 1.414 \text{ mm}\right] \tag{32}$$

Location	Internal Diameter	Design Pressure	Minimum Thickness
Fish Barrier	2.700 m	0.1788 MPa	1.414 mm
New Croton	2.200 m	0.3418 MPa	2.203 mm
Cochiti	3.700 m	0.3279 MPa	3.555 mm

Conclusion: These calculations established preliminary pressure-only wall thickness requirements for each location. Additional analysis of water hammer, corrosion, external loads, buckling, manufacturing requirements, and applicable safety standards is necessary before selecting a final penstock wall thickness.

3.3.5 Penstock Head Loss

4 Design Concepts

This chapter breaks down design concepts, and how they compile into our final design selection. We explain our functional decomposition, with all sub functions elaborated upon. Later, we break down our design process, from generating a list of viable dams, turbines, penstock gates, and fish passages, to ranking these based on selection criteria. Finally, we pull these together to present our final NPD site selection, along with the intended turbine, fish passage, and penstock gate combination.

4.1 Functional Decomposition

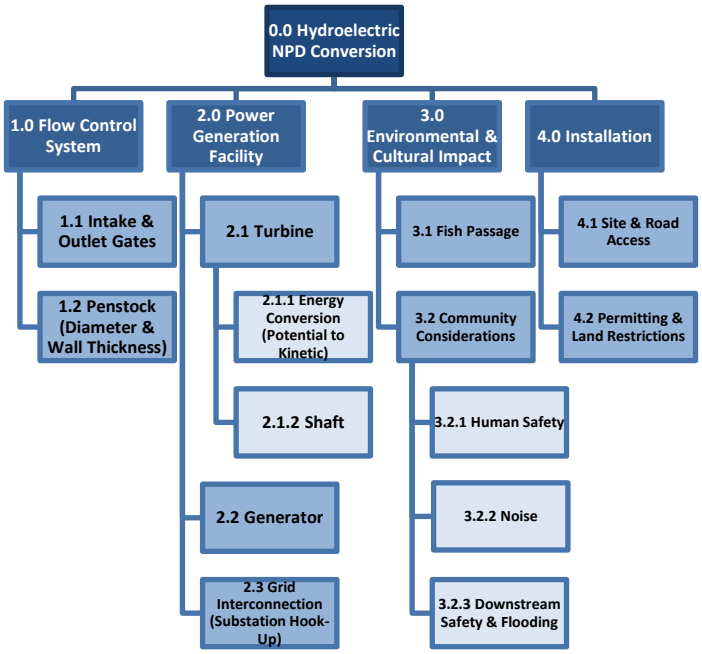


Figure 5: Functional Decomposition of hydroelectric NPD

4.1.1 Description

Figure 5 shows the functional decomposition chart for the team's non-powered dam (NPD) conversion. The team divides the hydropower plant into four systems: the flow control system, the power generation facility, environmental and cultural impacts, and installation. Each system represents a function the plant must perform: controlling flow, generating power, protecting the environment and community, and enabling construction. Each system then breaks down further into the subsystems the design must address.

4.1.2 Flow Control System (1.0)

The flow control system contains two subsystems. The first covers the intake and outlet gates, which control the flow of water into and out of the penstock and allow the plant to shut off flow when needed. The second subsystem is the penstock, the pipe that carries water from the reservoir to the turbine. The team sizes the penstock's diameter and wall thickness for the design flow and the pressure from the dam's head. Because the project converts an existing dam, the team first evaluates whether the dam's existing outlet works can connect to the penstock, since reusing them reduces cost and avoids cutting new openings in the dam.

4.1.3 Power Generation Facility (2.0)

The power generation facility forms the core of the project, and the team breaks it into three subsystems. The first is the turbine, which converts the energy of the water into mechanical energy. The dam's head sets how much potential energy the water carries, and the penstock turns that potential energy into the pressure and velocity that drive the turbine. The turbine's shaft then transfers this rotational energy to the generator, the second subsystem, which converts it into electrical energy. The team selects the turbine type from each candidate dam's head and sizes it is using flow duration curves built from 30 years of USGS data, choosing the design flow with the lowest levelized cost of energy. The final subsystem, the grid connection, links the plant to the nearest substation and delivers the power to the area that uses it. For this subsystem, the team evaluates each dam's distance to the nearest substation and whether that substation has the capacity to accept the plant's output.

4.1.4 Environmental and Cultural Impacts (3.0)

This system contains two subsystems. The first, fish passage, addresses fish migration and movement, which the local ecosystem depends on. The team compares fish passage options based on cost and dam height. The second, community considerations, covers human safety, noise, and downstream safety and flooding. The plant must operate without reducing the safety of the surrounding community or changing how the river behaves downstream. The team also aims to ensure the surrounding community benefits from the project and does not suffer from excessive noise or new power lines. The team limits these impacts by evaluating each dam's distance to residential areas.

4.1.5 Installation (4.0)

Installation contains two subsystems. The first, site and road access, determines whether crews and equipment can reach the dam for construction, and maintenance. The second, permitting and land restrictions, identifies who owns the dam and which permits the project requires. Both directly affect the cost and profitability of the plant. Because the project adds generation to a dam that already serves purposes such as flood control or water supply, the plant must not interfere with those uses.

4.1.6 Importance of the Decomposition

The decomposition gives the team a structured way to evaluate each dam's viability for conversion to a hydropower plant. Clear criteria matter because new information about a site can show that a dam is no longer feasible. Because each system and subsystem defines criteria that apply to any dam, the team can run the same evaluation on a new site instead of starting over. For example, if a site's actual head proves lower than the dataset reports, the team moves to the next ranked dam and repeats the same evaluation. This allows the team to pivot quickly to another candidate site for in-depth analysis and saves time when researching new sites.

The decomposition also links each system to the analysis that supports it. The flow duration curves and design-flow calculations size the flow control system, the dam's head determines the turbine type in the power generation facility, the fish passage study informs the environmental system, and the construction budget assigns a cost to every subsystem. This mapping confirms that the design addresses every requirement and assigns each team member a clear area of responsibility.

4.2 Concept Generation

4.2.1 Site Selection

Our concept generation centers on non-powered dams (NPDs) in the United States. Oak Ridge National Laboratory (ORNL) has identified roughly 2616 NPDs viable for conversion to hydropower (Figure 6) [22]. Their listed database provides these with estimates for power output, expected capacity, nominal flow, and other statistics that are necessary to determine conversion potential. Though this database is valuable in creating an initial selection of NPD conversion candidates, it fails to identify more nuanced considerations needed to identify the optimal dam for conversion. Certain aspects of site selection use information from the National Inventory of Dams (NID) [28], which compiled the specifications for NPDs in the United States, such as owners, usage, and structure.

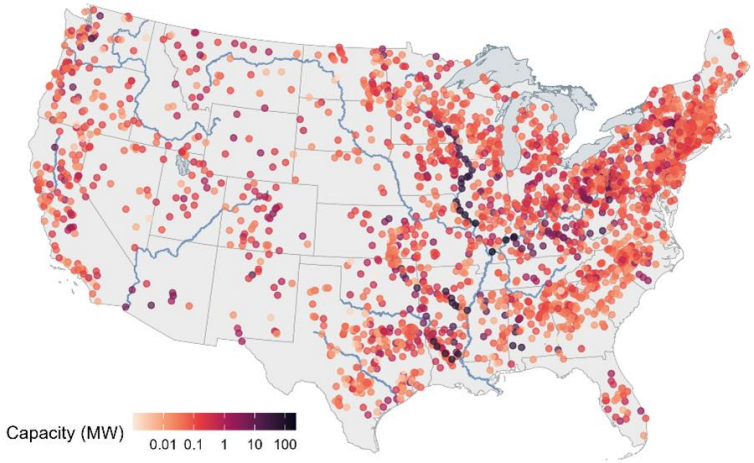


Figure 6: All NPD dams evaluated in the ORNL analysis for hydropower conversion

We created a custom MATLAB code, which filters these dams to eliminate dams with unfavorable conditions, either being outside of the project scope or determined as unfavorable by the team. These initial elimination criteria are nominal capacity, which must be within 1 – 10 MW, and being located within protected lands such as tribal lands or a national park. From our custom MATLAB code, the following three dams were identified as the top choices for dam conversion (site specifications listed in Table 3):

4.2.1.1 New Croton Reservoir Dam

The New Croton Reservoir Dam, located in Croton-on-Hudson, NY on the Croton River, is a high head, large reservoir structure with a listed mean head of 34.84 m and an average flow of 11.48 m^3/s. The structure is primarily concrete, and the outlet of the penstock empties into a generally flat area, which makes construction of a turbine retrofit more seamless. The dam is owned and operated by the New York City Department of Environmental Protection, a local government entity, and its listed purpose is water supply. ORNL estimates a capacity of 5.06 MW and a yearly output of 20,192 MWh, and the nearest substation is only 0.32 km away, the shortest distance of the three sites. The primary concern with the site is the age of the structure, as construction was completed in 1906. Older dams typically require more maintenance and possible large-scale renovations (depending on condition), posing higher risks to dam integrity.



Figure 7: Image of the New Croton Reservoir Dam

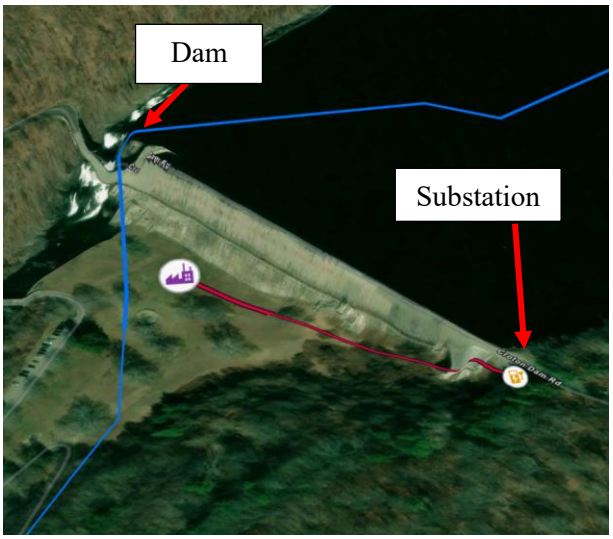


Figure 8: ArcGIS view of the New Croton Reservoir Dam

4.2.1.2 Little Falls Dam

The Little Falls Dam, located in Washington, DC on the Potomac River, is a low head, high flow run-of-river structure, meaning it passes the river's natural flow rather than storing a large reservoir. It has a listed mean head of only 2.4 m but the highest average flow of the three sites at 268.58 m^3/s. The structure is primarily concrete and already contains a fish passage solution, a weir-based fishway built around a notch at the top of the dam with a labyrinth of weirs within and downstream of the notch. The weirs reduce the velocity of the water flowing through the passage, allowing fish to swim upstream. The presence of an existing fish passage reduces overall retrofitting costs, which makes the dam more advantageous than those without. The dam is owned and operated by the U.S. Army Corps of Engineers, and its primary purpose is water supply, with additional unspecified applications listed. ORNL estimates the highest capacity and output of the three sites at 6.77 MW and 29,990 MWh/yr, with the substation 1.6 km away. The primary concern is the magnitude of head, with a listed head of 2.4 m, so any overestimate or underestimate in the database would strongly affect the expected output and whether turbine implementation is practical. We will need to complete further analysis into the real hydraulic head, to ensure that the site is operational at the minimum estimated head.



Figure 9: Image of the Little Falls Dam

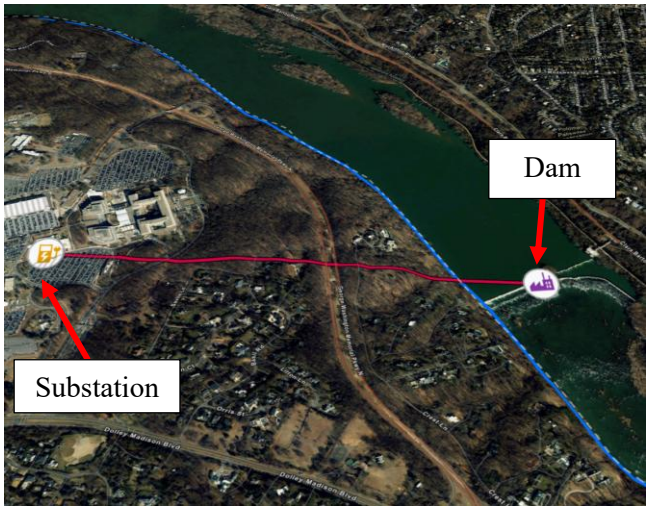


Figure 10: ArcGIS view of the Little Falls Dam

4.2.1.3 Birch Hill Dam

The Birch Hill Dam, located in Athol, MA on the Millers River, is a run-of-river structure with a moderate head of 6.2 m and an average flow of 17.33 m^3/s. The structure is primarily concrete and sits in an extremely rural area with few structures in the direct vicinity. It is owned and operated by the U.S. Army Corps of Engineers, and its listed purpose is Flood Risk Reduction. The remote location is ideal for limiting impact on daily operation and surrounding residents. However, the site has the lowest nominal capacity of the three at 1.21 MW, only slightly above the 1 MW project minimum, and a yearly output of 5,818 MWh. It is also the farthest from electrical infrastructure, with the nearest substation 4.2 km away, which raises the cost and losses of connecting to the grid.



Figure 11: Image of the Birch Hill Dam

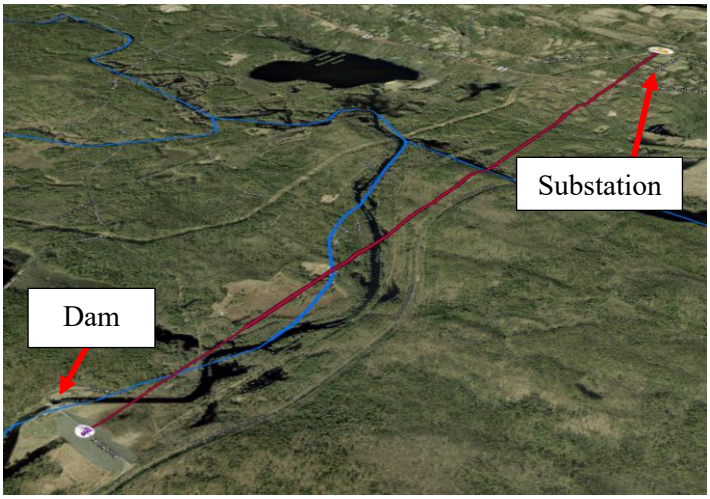


Figure 12: ArcGIS view of the Birch Hill Dam

4.2.2 Turbine Selection

From these dam selections, we can deduce the optimal turbine for the stream flow using the chart in Figure 26. For the project's constraints, to which the site's capacity must be within 1 – 10 MW, three turbines are feasible: Pelton, Francis, and Kaplan. Pelton is best for high head, low flow conditions, Francis is best for medium head, medium flow conditions, and Kaplan is best for low head, high flow conditions. Table 4 displays the operating conditions for each turbine, including its efficiency and minimum operating flow.

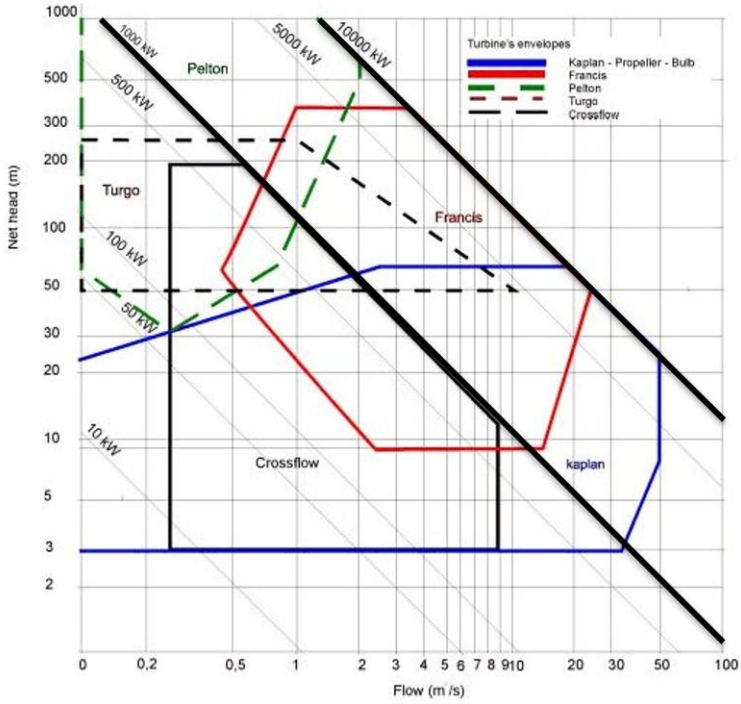


Figure 13: Optimal turbine for stream flow conditions [26]

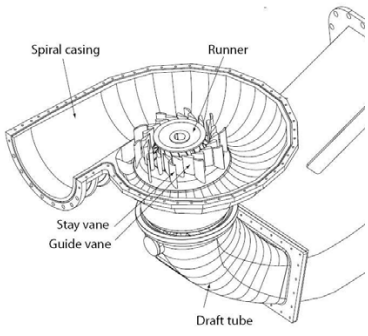


Figure 14: Francis Turbine



Figure 15: Pelton Turbine



Figure 16: Kaplan Turbine

4.2.3 Fish Passage Selection

One of the largest environmental impacts of dams is the disruption of fish pathways in the ecosystem, so a clear route for fish to continue unobstructed is needed. Each passage option has advantages and disadvantages based on typical application and performance. For this analysis, three options were evaluated: a nature-like bypass, a fish lift, and a vertical-slot fishway [15], [16].

4.2.3.1 Nature-Like Bypass

A nature-like bypass offers the benefit of serving as additional habitat for aquatic life, and it requires minimal concrete and construction cost. However, it has a low attraction rate of 48%, requires very low slopes, and its performance varies by site.

4.2.3.2 Fish Lift

The fish lift is the preferred choice for dams over 25 ft, which is its main advantage. Its drawbacks are that fish is harder to attract to the lift and that its operation is not consistent.

4.2.3.3 Vertical-Slot Fishway

The vertical slot fishway has a strong attraction rate, with 80% of fish finding the passage, and it works well with wide slots and with low slopes or lower dams. However, passage success once fish are inside low at 45%, and the narrow slots force fish to move in single file.

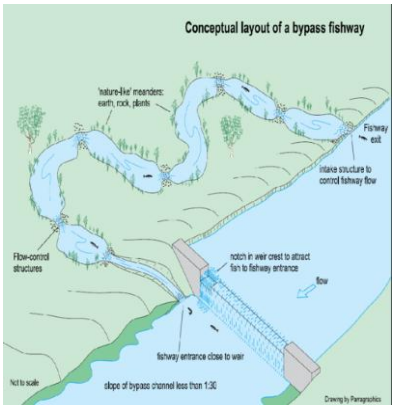


Figure 17: Nature-Like Bypass



Figure 18: Fish Lift

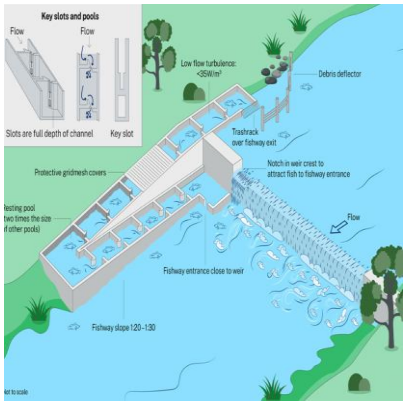


Figure 19: Vertical-Slot Fishway

4.2.4 Penstock Gate Selection

One of the key design decisions for a dam is choosing a gate that can reliably control or shut off flow, including under emergency conditions. Each gate option has advantages and disadvantages based on typical application and performance. For this analysis, three options were evaluated: a fixed wheel vertical lift gate, a tractor gate, and a high pressure hydraulic slide gate.

4.2.4.1 Fixed Wheel Vertical Lift Gate

The fixed wheel vertical lift gate is suitable for large openings, and its gravity closure has proven successful for emergency use. However, it needs a lot of space and large guides, it requires maintenance, and debris can affect the seal.

4.2.4.2 Tractor Gate

The tractor gate distributes loading evenly along its guides, which makes it a good choice for heavily loaded openings, and it can serve as both an emergency stop and a regular closure. Its disadvantages are that it has many moving parts, needs proper maintenance access, and is prone to corrosion.

4.2.4.3 High Pressure Hydraulic Slide Gate

The high pressure hydraulic slide gate is good for closing against flow and uneven head, has fewer moving parts, and offers positive hydraulic actuation. However, sliding friction causes more wear on the seals, and it is not a good option for large penstock openings.



Figure 20: Fixed Wheel Vertical Lift Gate



Figure 21: Tractor Gate



Figure 22: High Pressure Hydraulic Slide Gate

4.3 Selection Criteria

To rank the dams that passed screening, we scored each one on five weighted criteria using the custom MATLAB program. From our research, we used the following metrics:

1. Theoretical Yearly Power Generation: More annual energy output means a larger return on the conversion, so higher is better. We used ORNL's estimate of yearly generation from NPD conversion potential, which assumes the dam is converted using its measured flow and head.
2. Flow and Head Variance (separate criteria): Variability in flow and head leads to variability in output, and an uncertain output makes a project harder to justify financially. To quantify this, we calculated the coefficient of variation of the monthly flow and monthly head values in the ORNL dataset for each dam, with lower variance scoring higher.
3. Energy Loss: Power lost in transmission between the generator and the grid reduces the energy a dam can deliver, so lower loss is better. We estimated loss for each dam from its expected power output and its distance to the nearest substation, using substation locations from OpenStreetMap. This rewards sites that sit close to existing grid infrastructure.
4. Energy Need: A community with high electricity costs likely has either high demand or limited supply, so a new local energy source would be more valuable there. We used NREL data on residential energy cost for each dam's ZIP code, covering both private investor-owned and public municipal utilities. A higher local rate scored higher.

After these criteria were applied to the remaining dams, 209 dams of the original 2616 dams were ranked using weights determined by a Pairwise matrix (Table 5). The highest scoring sites were carried forward for closer review. The following considerations were considered to reduce the top-ranked list:

- Head Magnitude: The ORNL database is a mass evaluation built from general stream data, so its head values may not match conditions at a given site. Some sites with low head in practice may be listed with much higher values, which inflates their expected output. Examples include Fish Barrier Dam in Washington, shown in Figure 23, where the listed mean head of 25.8 ft is unlikely, given images of the dam itself. These sites need to be checked individually against site data and image references rather than trusted from the ranking alone.
- Dam Type: Similar to the head magnitude dilemma in the ORNL database, the same issue arises with the dam type. Some dams, such as Lock E10 Cranesville in New York, shown in Figure 24, are moveable dams. These dams are impossible to convert to hydropower, as the dam must be a solid, immovable structure to which a turbine can be fixed to. Sites like this one must be removed.
- Residential Impact: We treated this as a soft elimination. The effect of a project on a surrounding community cannot be properly quantified with the data sets available to us. Our current view is that it is not a deal breaker, but it is not prioritized in the ranking. We plan to revisit sites near residential areas if most of the top ranked dams turn out to be unreasonable for other reasons.



Figure 23: Image of Fish Barrier Dam, WA (listed mean head: 25.8 ft)

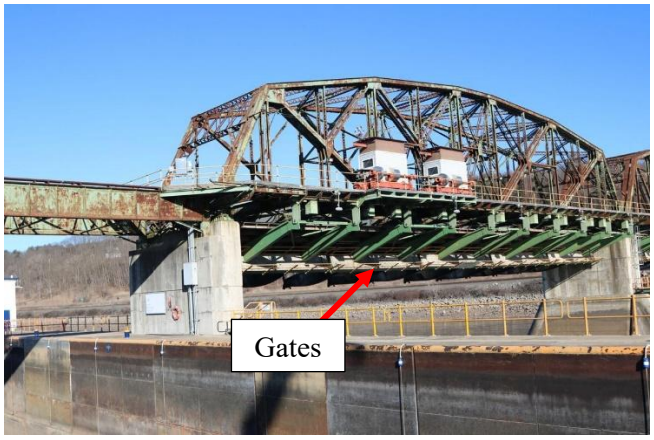


Figure 24: Image of Lock E10 Cranesville, NY (green gates open for boat passage)

4.4 Concept Selection

The following decision matrix shown in Table 2 shows the weighted rankings of each selected dam. From our analysis, New Croton Reservoir Dam was chosen as the optimal site for conversion. Given the site's conditions, referring to the turbine selection mathematical modelling, the optimal turbine configuration for the site is a Francis turbine. The best option for penstock gate at this site is the fixed-wheel vertical lift gate, as the system's rolling supports reduce friction, which aids operation under large hydraulic loads. Given that fish lifts are generally preferred for heads above 25 ft of head, it is the best choice for this configuration.

Table 2: Decision matrix of the top three dam sites

		New Croton Reservoir Dam		Little Falls Dam		Birch Hill Dam	
Criterion	Weight	Score out of 1	Weighted Score	Score out of 1	Weighted Score	Score out of 1	Weighted Score
Estimated Power Output	21.7%	0.47	0.102	0.72	0.156	0.09	0.020
Flow Rate Variance	9.0%	0.47	0.042	0.46	0.041	0.49	0.044
Head Variance	22.5%	0.92	0.207	0.92	0.207	0.88	0.198
Energy Loss (Transmission)	24.9%	0.99	0.247	0.92	0.229	0.96	0.239
Energy Need	21.9%	0.90	0.197	0.29	0.064	0.89	0.195
Total	100%		0.79		0.70		0.70
Rank		1		2		3	

4.4.1 Final Site Selection



Figure 25: New Croton Reservoir Dam

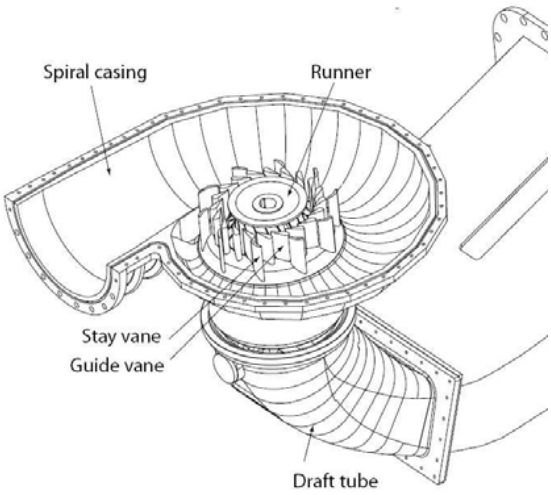


Figure 26: Francis Turbine



Figure 27: Fixed Wheel Gate



Figure 28: Fish Lift

5 CONCLUSIONS

The purpose of this project is to evaluate the feasibility of converting existing non powered dams into hydropower facilities capable of producing between 1 and 10 MW of electricity. The design must consider power generation, installation, site conditions, environmental impacts, and practical design limitations as it pertains to the existing infrastructure. During this semester the team researched extensively into established engineering requirements, developed mathematical models and evaluated potential sites and preliminary design concepts.

The site selection model allowed the team to compare approximately 2600 candidate dams based on power generation potential, flow and head variability, proximity to electrical infrastructure, and local energy demand. This analysis narrowed the initial dataset to a smaller set of promising sites selected for further evaluation and verification for feasibility. The results demonstrated the delicate balance of placing importance on a variety of variables in order to identify suitable hydropower conversion locations.

Further work is needed to confirm that the proposed system is feasible. The team must verify actual site conditions and data, then refine turbine design flow, penstock geometry and dimensions, evaluate losses within the system, structural requirements, and electrical connections. Installation and operating cost estimates will also need to be further refined and additional research is needed to account for fish passage, permitting, and potential effects on existing dam operations. Overall this semester's work established a strong method to decide site selection, and a preliminary system concept. The next phase will focus on validating current assumptions and refining the design to determine if it will meet the project's technical, economic, safety, and environmental requirements.

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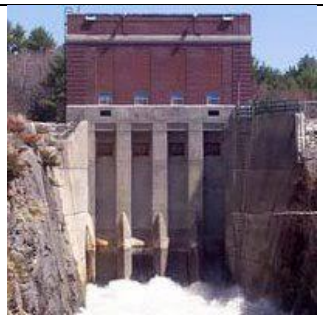
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7 APPENDICES

7.1 Appendix A: Selection Resources

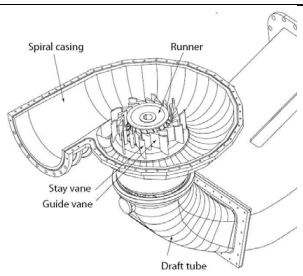
7.1.1 Concept Generation: Dam Selection

Table 3: Top-rated dam sites for hydropower conversion

	NEW CROTON RESERVOIR DAM	**LITTLE FALLS DAM**	**BIRCH HILL DAM**
			
Capacity (MW)	5.06	6.77	1.21
Output (MWh/yr)	20192	29990	5818
Average Flow (m^3/s) & CV	11.48 / 0.64	268.58 / 0.65	17.33 / 0.65
Head (m) & CV	34.84 / 0.04	2.4 / 0.04	6.2 / 0.05
Distance to Substation (km)	0.32	1.6	4.2
Utility Energy Cost ($/kWh)	0.29	0.15	

7.1.2 Concept Generation: Turbine Selection

Table 4: Selected turbine options given project specifications

	FRANCIS TURBINE	PELTON TURBINE	KAPLAN TURBINE
			
Type	Reaction: fully submerged runner driven by water pressure [6]	Impulse: a water jet strikes buckets in open air [6]	Reaction: fully submerged propeller-style runner [6]
Head	25-350 m [6],[7]	50-1300 m [6],[7]	2-40 m [6],[7]
Flow rate	Medium, 0.45- 25 $\frac{m^3}{s}$ [6]	Low 0.02-2.5 $\frac{m^3}{s}$ [6]	High 0.3-50 $\frac{m^3}{s}$ [6]
Peak efficiency	0.94 [6]	0.89-0.90 [6]	0.93 [6]
Minimum operating flow	50% of design flow [6],[7]	10% of design flow [6],[7]	15% of design flow [6],[7]
Specific speed	0.05-0.33 [7]	0.005–0.025 [7]	0.19–1.55 [7]

7.1.3 Concept Evaluation: Pairwise Matrix

Table 5: Pairwise matrix for determining the weight of each selection criteria

Weights for MATLAB code							
	Variance of Head	Variance of Flow	Infrastructure Distance	Power Generation	Local Energy Cost	Geometric Mean	Weights
Variance of Head	1.000	2.000	1.500	0.800	1.000	1.191	0.225
Variance of Flow	0.500	1.000	0.300	0.330	0.500	0.477	0.090
Infrastructure Distance	0.667	3.333	1.000	1.500	1.200	1.320	0.249
Power Generation	1.250	3.030	0.667	1.000	0.800	1.151	0.217
Local Energy Cost	1.000	2.000	0.833	1.250	1.000	1.158	0.219

7.2 Appendix B: Flow Duration Curves and Design Flow Analyses

7.2.1 Birch Hill Dam

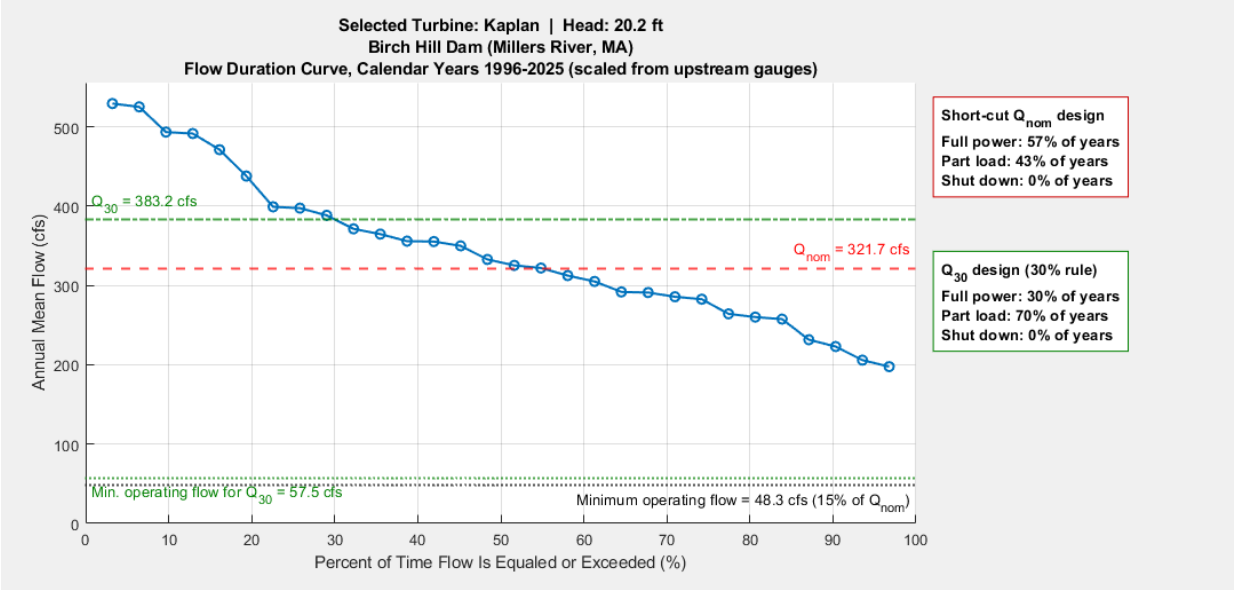


Figure 29: Birch Hill Dam Flow Duration chart.

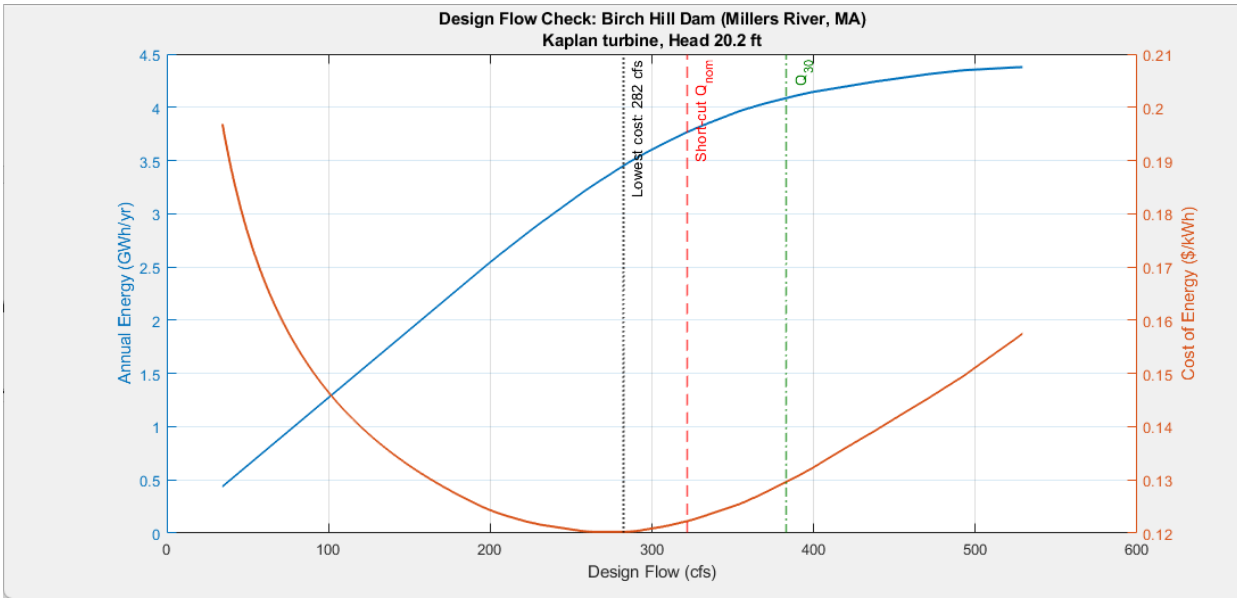


Figure 30: Birch Hill Dam -power output and energy cost at specific design flows

7.2.2 Little Falls Dam

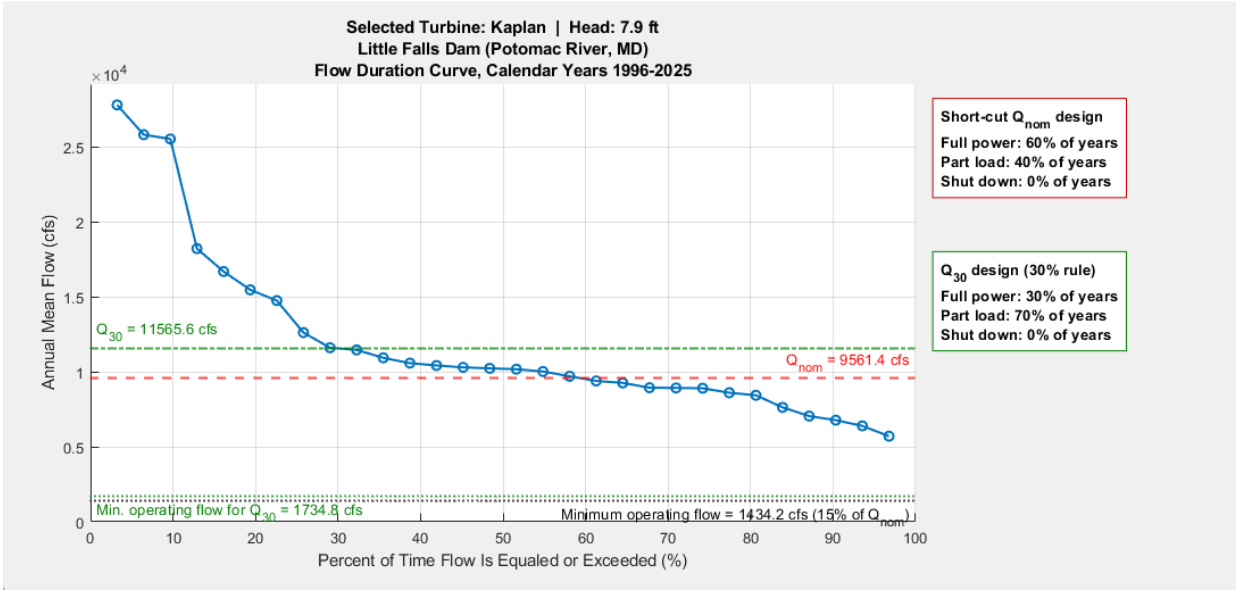


Figure 31: Little Falls Dam Flow Duration chart

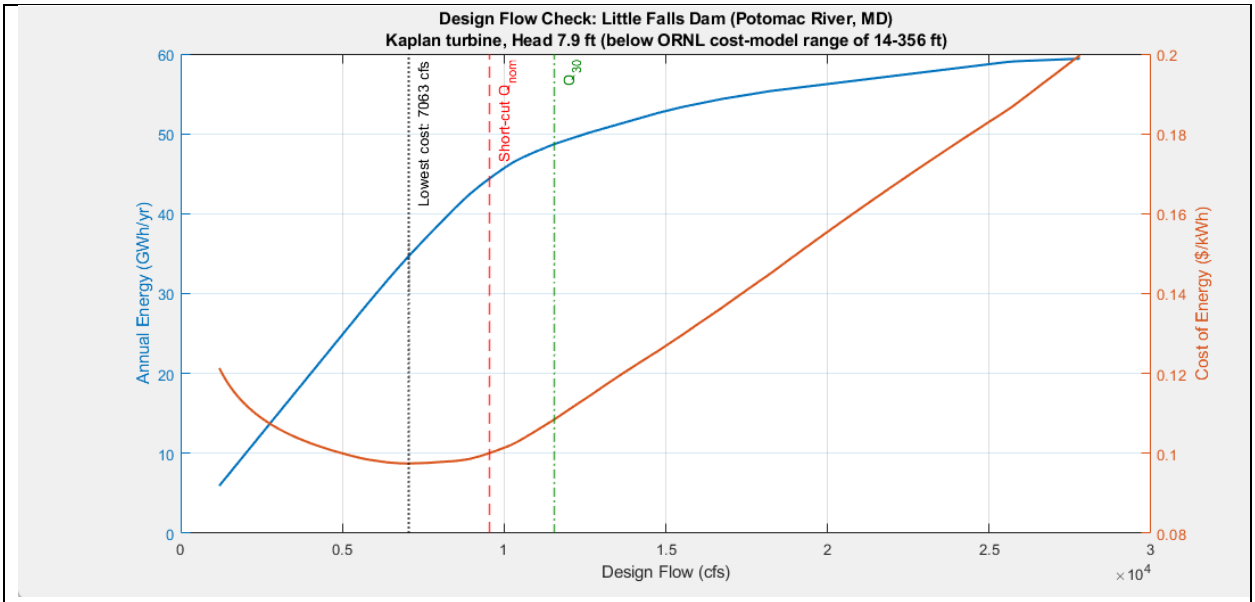


Figure 32: Little Falls Dam -power output and energy cost at specific design flows

7.3 Appendix C: Charts for Efficiency Estimation

7.3.1 Moody Chart

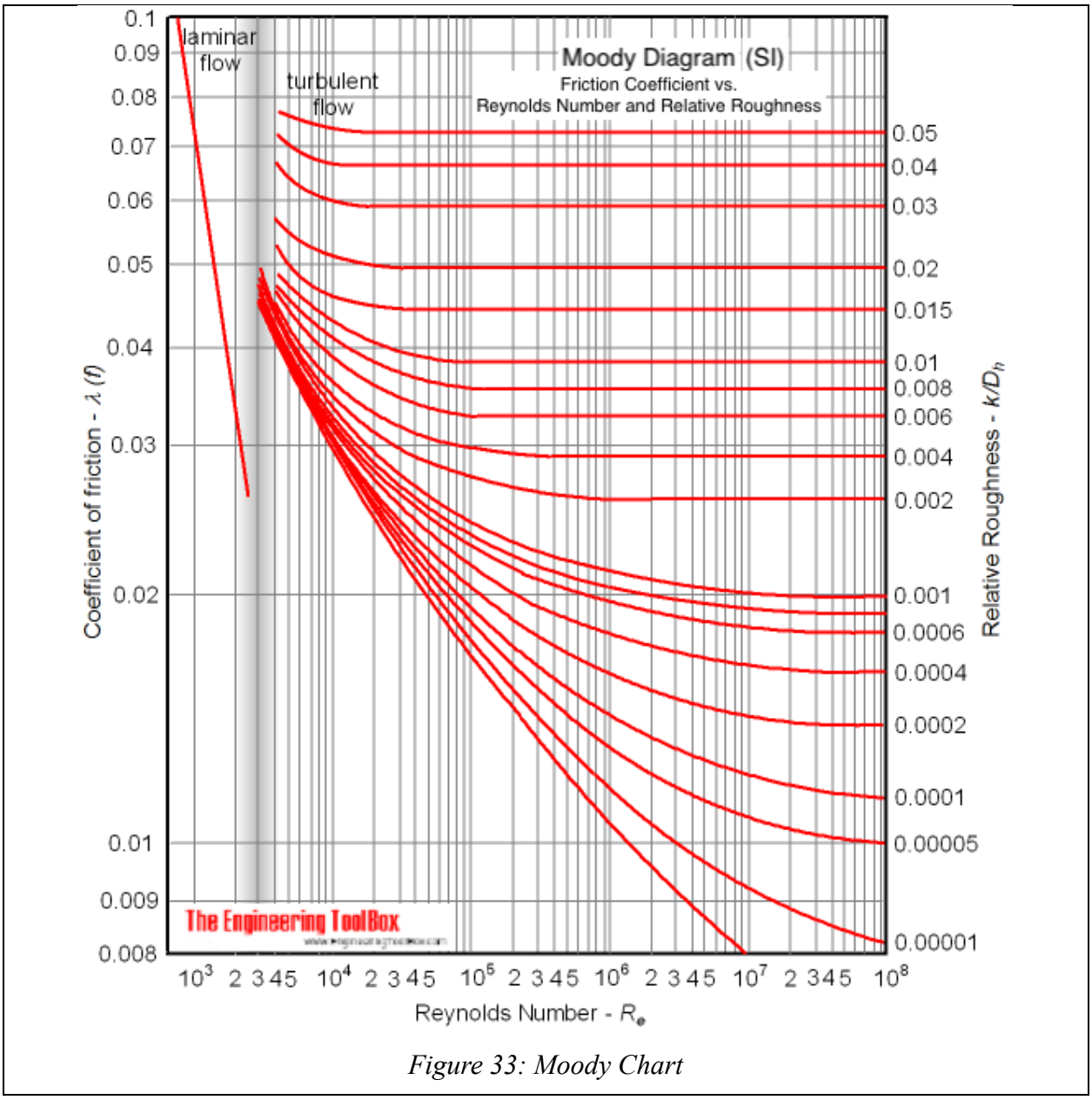


Figure 33: Moody Chart

7.3.2 Turbine Efficiency Curves

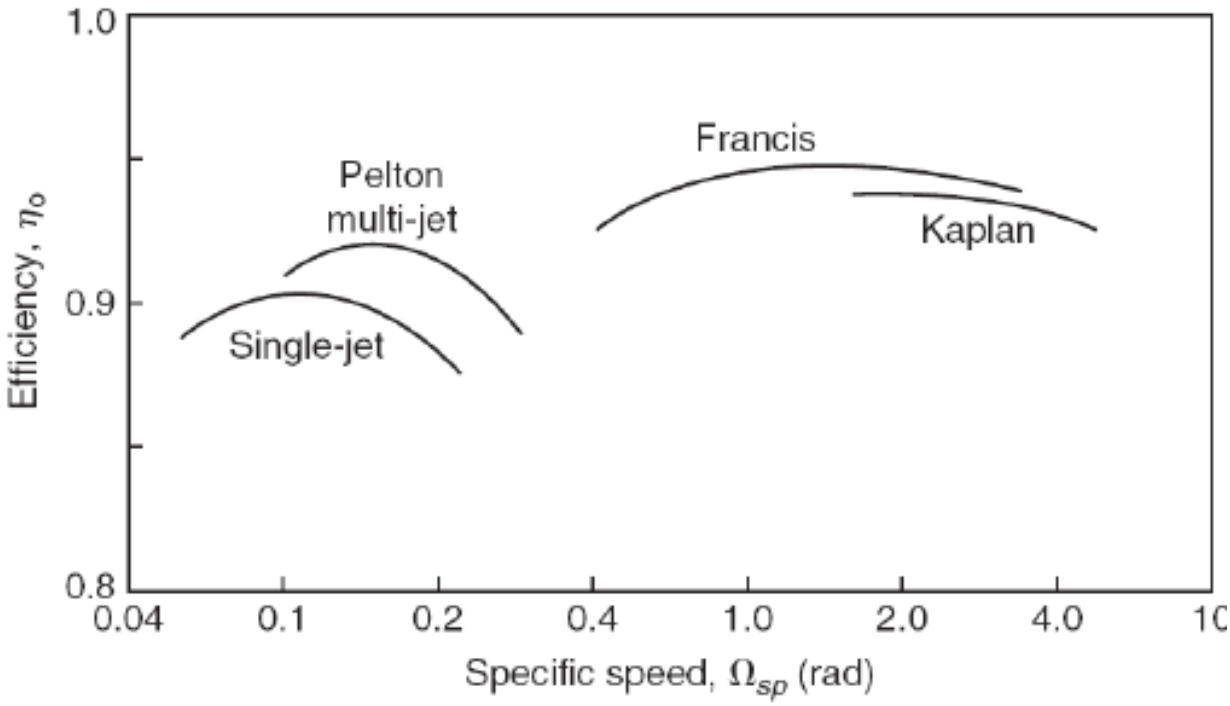


Figure 34: Efficiency curve graphs (specific speed vs efficiency)